Set 1C: Stages Coupled by Dependent Sources

ECEN 325 — Lecture 1 (Linear Network Analysis) practice set

What this set trains. Circuits split into stages by dependent sources (diamond symbols). A VCVS (kV_x) forces its output voltage; a VCCS (gV_x) pushes a current into the impedance on its node. Because a dependent source does not load the node it watches, $H(s)$ is simply the product of the stage gains, and each stage is solved with voltage dividers. Problems 1–6 are easy, 7–12 medium, 13–18 hard. Try each problem before looking at its solution.

The Lecture-1 pipeline (use it for every problem in this set)

Tools — this is all the theory you need:

T1 *Impedances:* $Z_R = R$, $Z_C = \frac{1}{sC}$, $Z_L = sL$. Series: add. Parallel: $Z_a \parallel Z_b = \frac{Z_a Z_b}{Z_a + Z_b}$.

T2 *Voltage divider:* if Z_{top} connects V_{in} to a node and Z_{bot} is everything from that node to ground, then $V_{node} = V_{in}\,\frac{Z_{bot}}{Z_{top}+Z_{bot}}$.

T3 *Divider between two voltages (DC):* a node tied to V_a through R_a and to V_b through R_b sits at $V = V_b + (V_a - V_b)\frac{R_b}{R_a+R_b}$, and looks like a source V with resistance $R_a \parallel R_b$ (its Thevenin pair).

T4 *Dependent sources:* a VCVS forces $V = kV_x$; a VCCS drives the current gV_x through the impedance Z hanging on its node, so $V = \pm gV_x Z$ (sign from the arrow).

T5 *Sinusoids:* $A\sin(\omega t + \theta) \;\rightarrow\; |H(j\omega)|\,A\sin\big(\omega t + \theta + \angle H(j\omega)\big)$.

T6 *Bode rules:* a pole at p bends the magnitude by $-20\,\text{dB/dec}$ above p, a zero by $+20\,\text{dB/dec}$; $\frac{1}{1+s/p}$ is $0\,\text{dB}$ below p; $\frac{s}{s+p}$ is $0\,\text{dB}$ above p.

Steps:

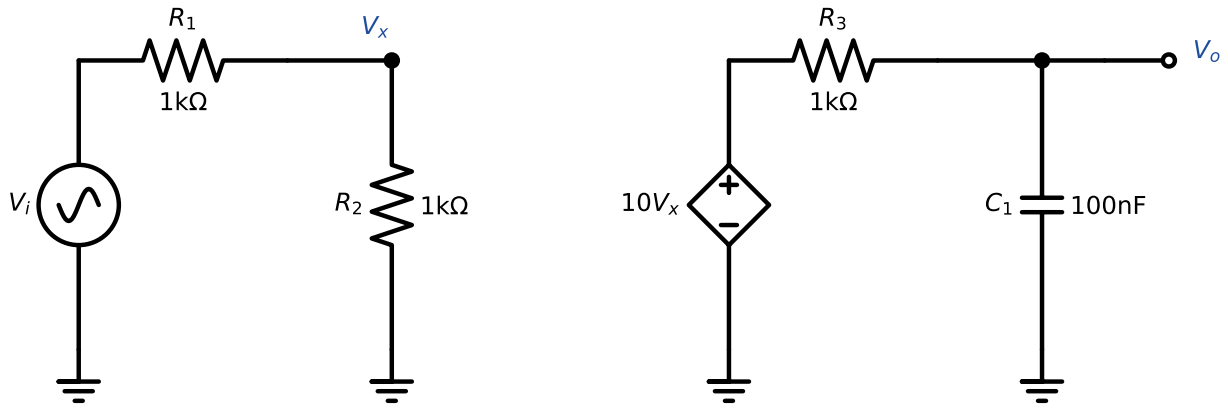
1. *Sources.* DC and AC both present $\Rightarrow$ superposition $V_o = V_{o,dc} + v_{o,ac}$.
2. *DC circuit:* $V_i \to 0$, $C \to$ open, $L \to$ short; reduce toward V_o with T3.
3. *AC circuit:* DC supplies $\to 0$ (rails become ground); merge parallel resistors.
4. *Divider chain:* name the nodes from V_i to V_o; $H = \prod$ (node-to-node ratios), each by T2 or T4, using the impedance seen looking right from each node.
5. *Standard form:* simplify, find the roots, write $H = K \cdot \prod(\text{factors})$.
6. *Bode:* start from the low-frequency slope, bend by $\pm 20\,\text{dB/dec}$ at each corner.
7. *Output:* evaluate each factor at $s = j\omega_{in}$, multiply magnitudes, add angles, apply T5 and add $V_{o,dc}$.

Problems

Easy problems

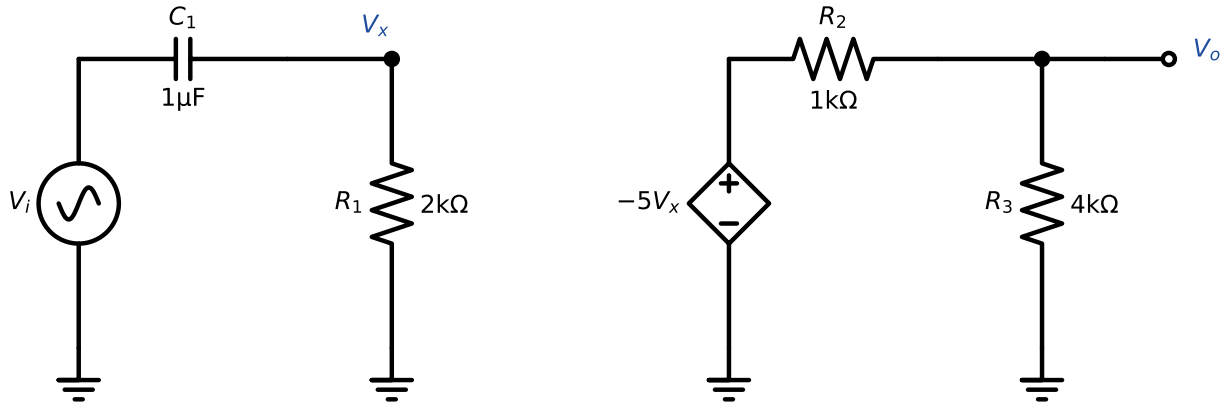
Problem 1C.1 — Easy

For the circuit below: (a) find the AC transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.1\sin(2\pi \cdot 1000\,t)$.



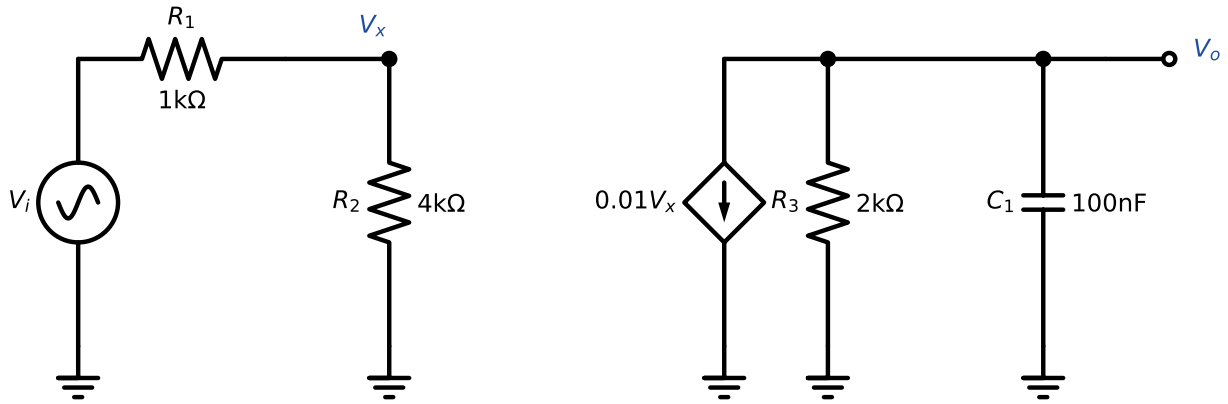
Problem 1C.2 — Easy

For the circuit below: (a) find the AC transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.2\sin(2\pi \cdot 100\,t)$.



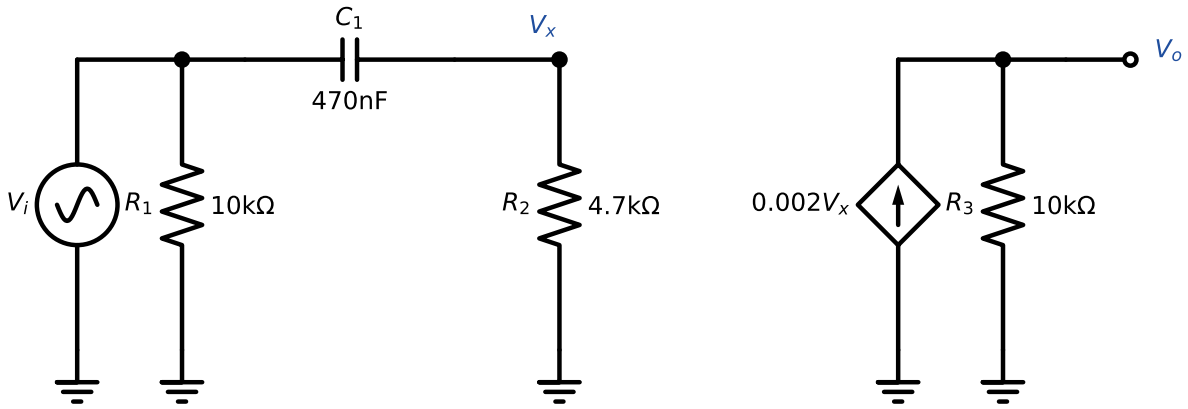
Problem 1C.3 — Easy

For the circuit below: (a) find the AC transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.05\sin(2\pi \cdot 500\,t)$.



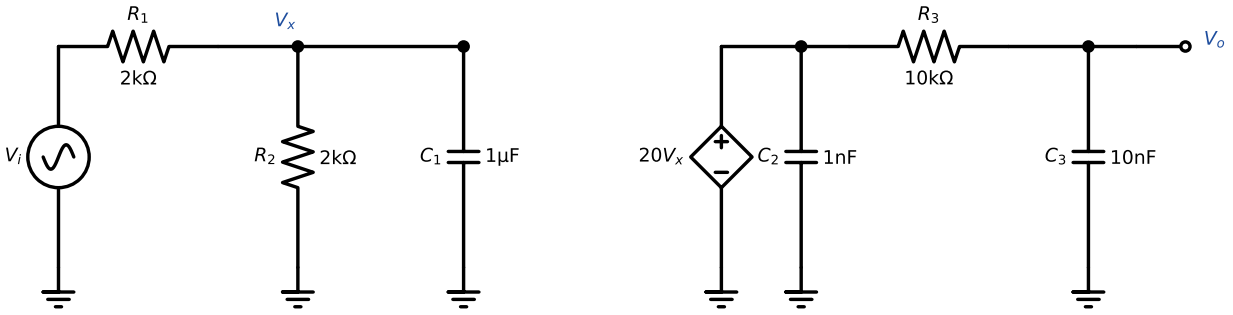
Problem 1C.4 — Easy

For the circuit below: (a) find the AC transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.1\sin(2\pi \cdot 200\,t)$.



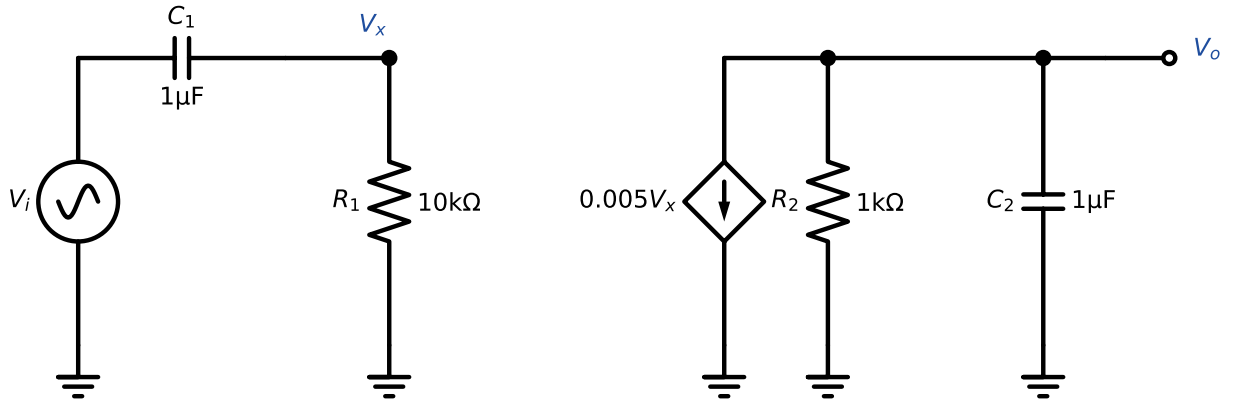
Problem 1C.5 — Easy

For the circuit below: (a) find the AC transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.05\sin(2\pi \cdot 300\,t)$.



Problem 1C.6 — Easy

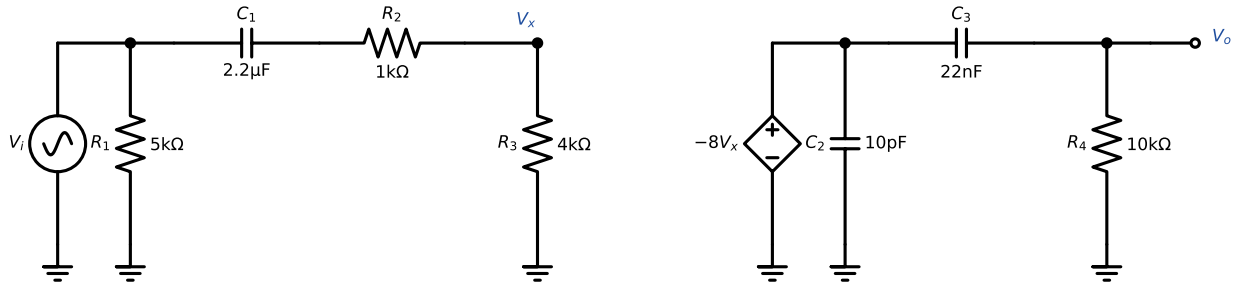
For the circuit below: (a) find the AC transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.2\sin(2\pi \cdot 50\,t)$.



Medium problems

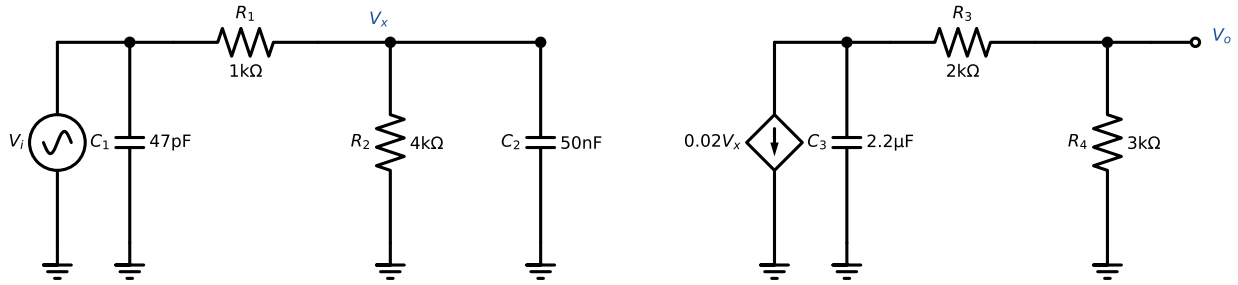
Problem 1C.7 — Medium

For the circuit below: (a) find the AC transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.1\sin(2\pi \cdot 1000\,t)$.



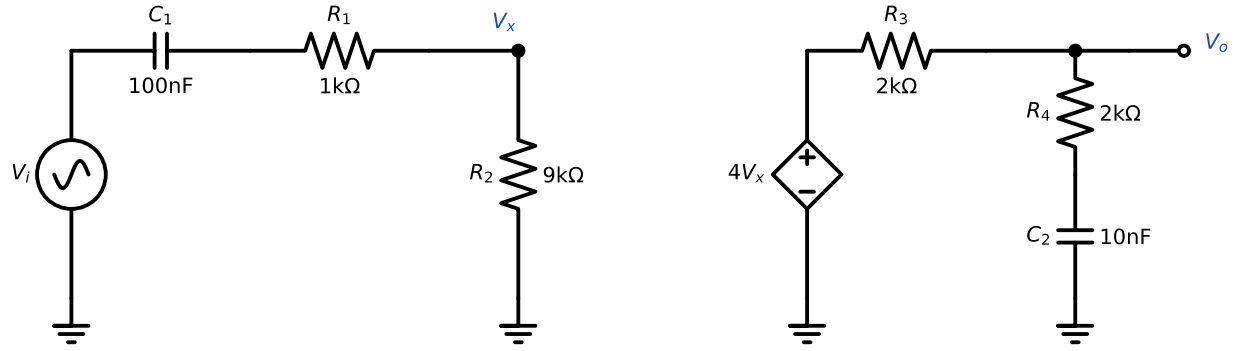
Problem 1C.8 — Medium

For the circuit below: (a) find the AC transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.05\sin(2\pi \cdot 2000\,t)$.



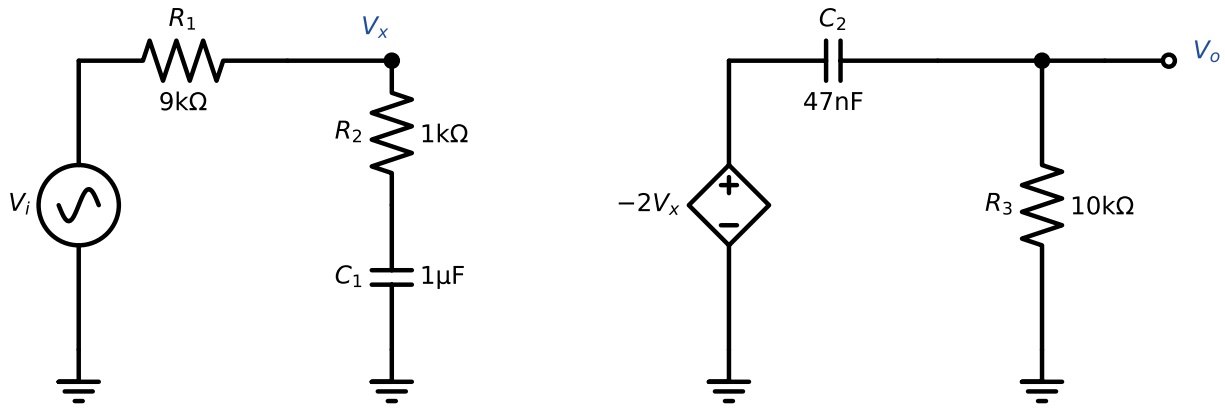
Problem 1C.9 — Medium

For the circuit below: (a) find the AC transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.2\sin(2\pi \cdot 5000\,t)$.



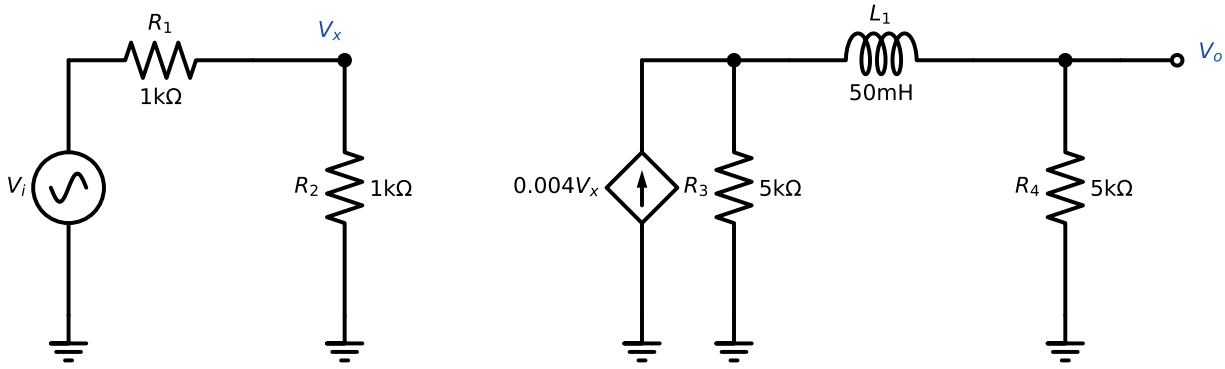
Problem 1C.10 — Medium

For the circuit below: (a) find the AC transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.5\sin(2\pi \cdot 100\,t)$.



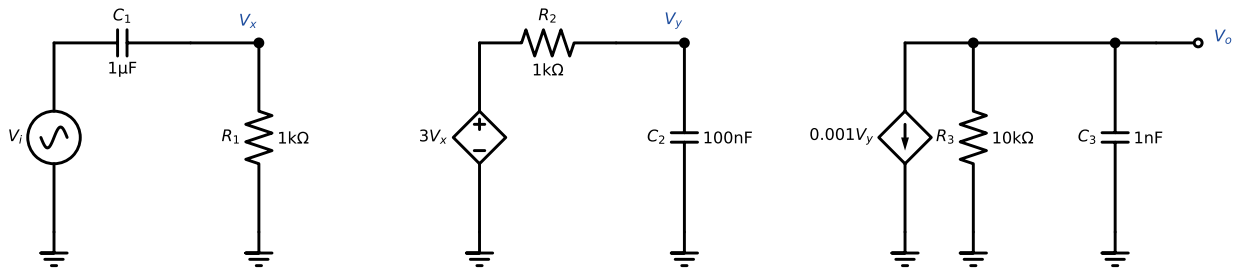
Problem 1C.11 — Medium

For the circuit below: (a) find the AC transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.2\sin(2\pi \cdot 20000\,t)$.



Problem 1C.12 — **Medium**

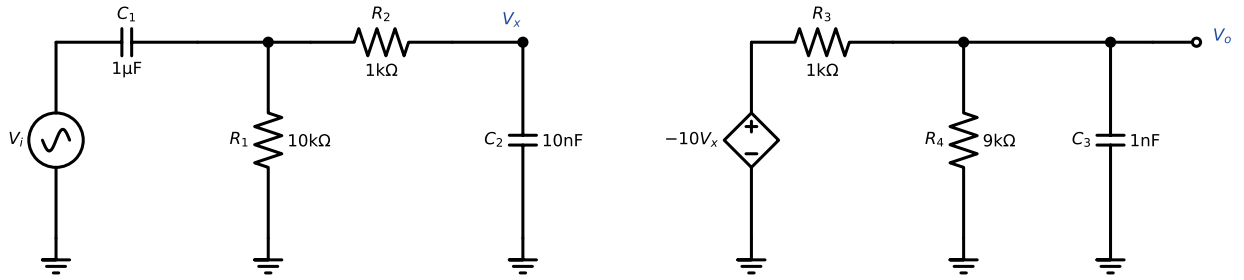
For the circuit below: (a) find the AC transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.1\sin(2\pi \cdot 1000\,t)$.



Hard problems

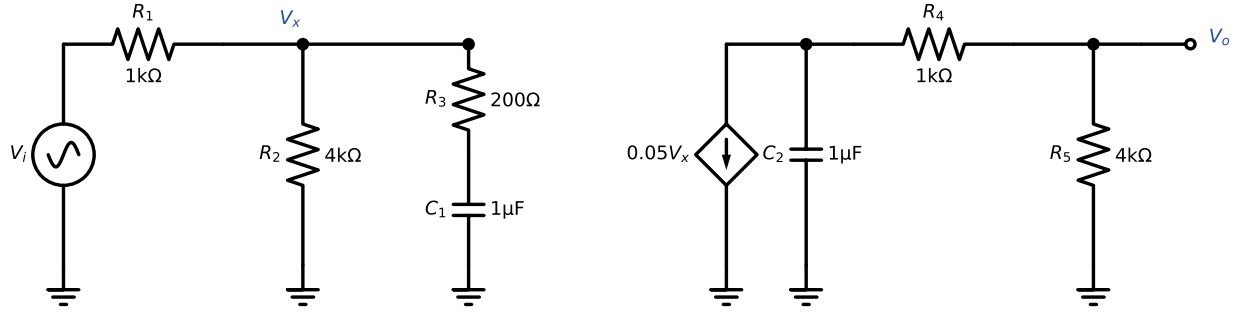
Problem 1C.13 — **Hard**

For the circuit below: (a) find the AC transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.05\sin(2\pi \cdot 2000\,t)$.



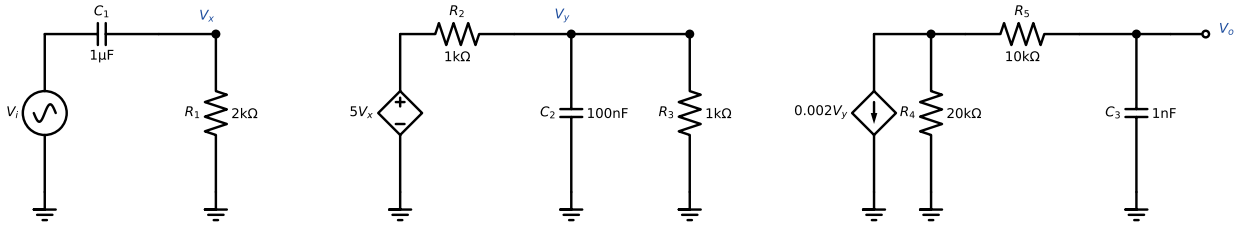
Problem 1C.14 — **Hard**

For the circuit below: (a) find the AC transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.02\sin(2\pi \cdot 1000\,t)$.



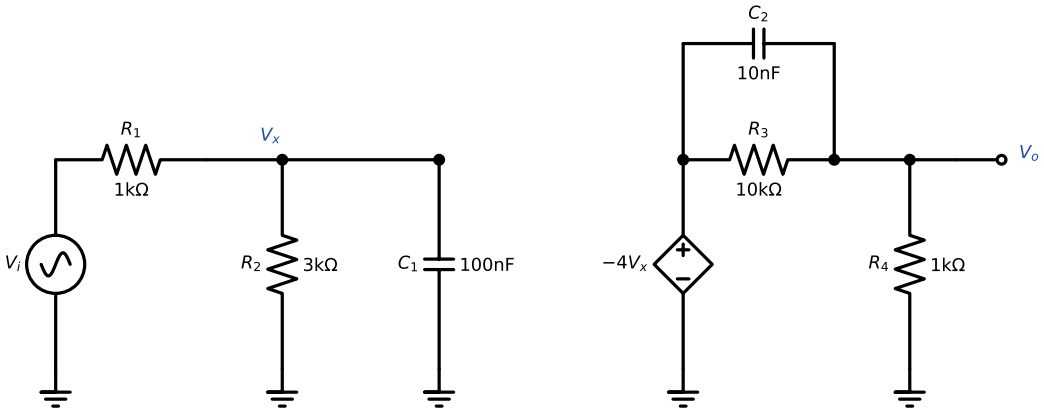
Problem 1C.15 — Hard

For the circuit below: (a) find the AC transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.1\sin\left(2\pi \cdot 2000\,t + \frac{\pi}{4}\right)$.



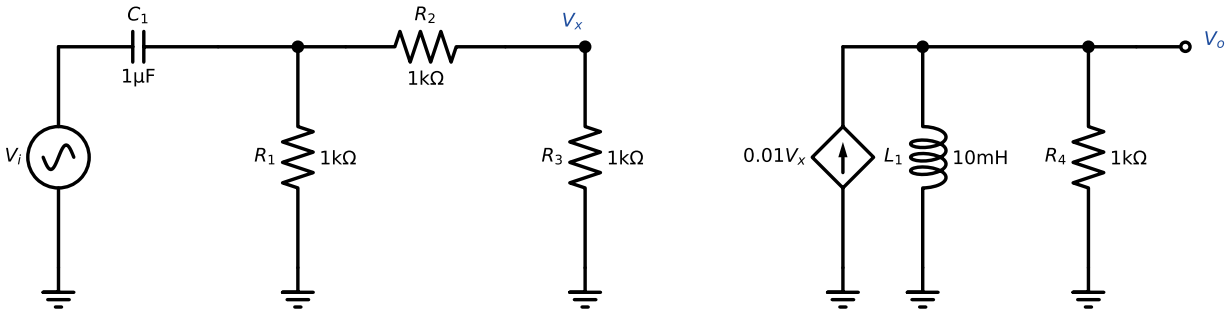
Problem 1C.16 — Hard

For the circuit below: (a) find the AC transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.1\cos\left(2\pi \cdot 3000\,t\right)$.



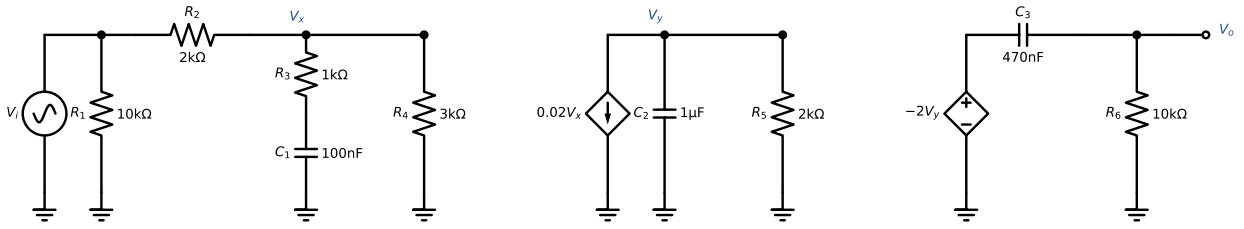
Problem 1C.17 — Hard

For the circuit below: (a) find the AC transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.2\sin\left(2\pi \cdot 5000\,t - \frac{\pi}{6}\right)$.



Problem 1C.18 **Hard**

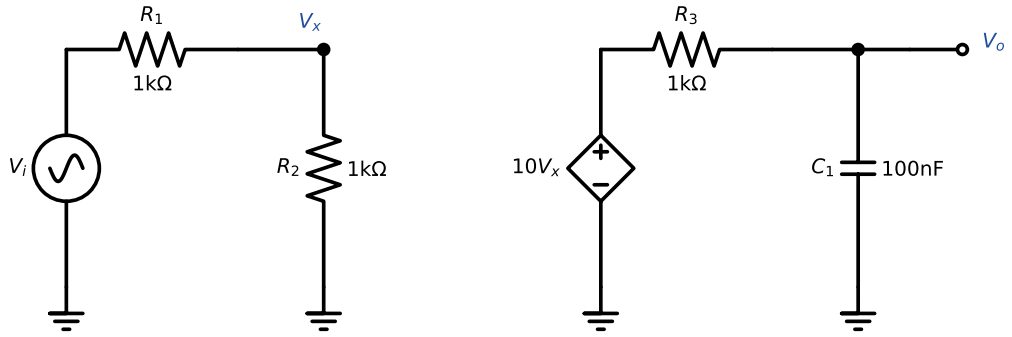
For the circuit below: (a) find the AC transfer function $H(s) = \dfrac{V_o(s)}{V_i(s)}$ and sketch its magnitude Bode plot; (b) find the steady-state output $V_o(t)$ for $V_i(t) = 0.01\cos\left(2\pi \cdot 500\, t + \frac{\pi}{3}\right)$.



Solutions

Every solution follows the same seven steps of the pipeline card.

Solution 1C.1 **Easy**



Step 1 — Identify the sources

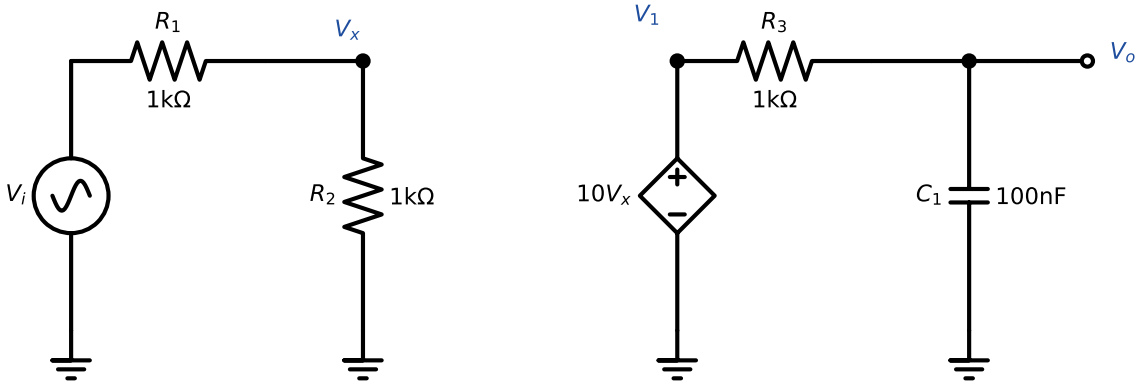
Independent sources: the AC input $V_i(t)$. The diamond-shaped sources are *dependent* sources: they are not inputs, they only copy/scale a voltage that already exists in the circuit, so they stay in the circuit in every step. There is no DC source, so $V_{o,dc} = 0$ and $V_o(t) = v_{o,ac}(t)$: Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — AC circuit

Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL. AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_x \to V_1 \to V_o$. The transfer function is the product of the node-to-node ratios (each one is a simple divider, tool T2, or a dependent source, tool T4):

$$H(s) = \frac{V_o}{V_i} = \frac{V_x}{V_i} \cdot \frac{V_1}{V_x} \cdot \frac{V_o}{V_1}$$

Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_x}{V_i} = \frac{R_2}{R_1 + R_2} = 0.5$$

$$\frac{V_1}{V_x} = 10 \qquad \text{(ideal VCVS: } V_1 = 10V_x\text{)}$$

$$\frac{V_o}{V_1} = \frac{\frac{1}{sC_1}}{R_3 + \frac{1}{sC_1}} = \frac{10000}{s + 10000}$$

Multiply the links and simplify (common factors cancel):

$$H(s) = \frac{50000}{s + 10000}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Denominator: $s + 10000$; $s + 10000 = 0 \Rightarrow s = -10000$.

Factored form (leading coefficient in front):

$$H(s) = 50000\,\frac{1}{(s + 10000)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 5\,\frac{1}{1 + \dfrac{s}{2\pi \cdot 1.59\text{k}}}, \qquad K = 5$$

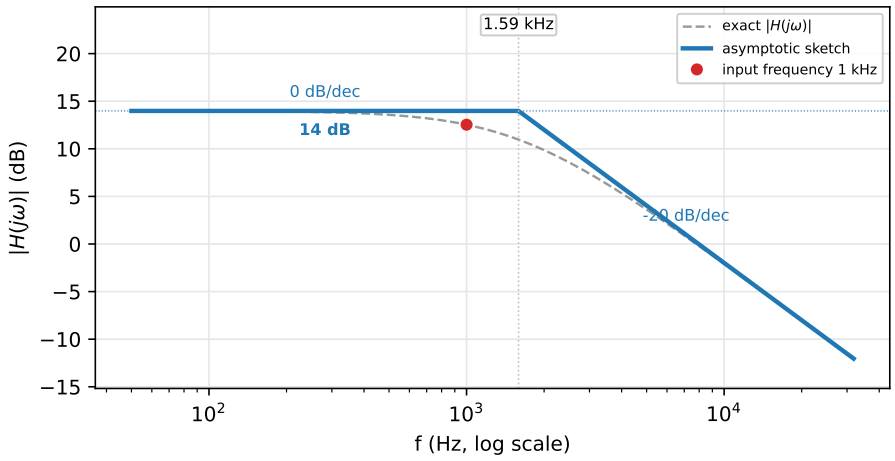
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	10000	1.59 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 5 = 14\,\text{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 1.59$ kHz	0 dB/dec (flat)	14 dB at end
$f > 1.59$ kHz	−20 dB/dec	14 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10} 5 = 14\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.1\sin(2\pi \cdot 1000\,t)$, so $\omega_{in} = 2\pi \cdot 1000 = 6280\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
5	5	0
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 1.59\text{k}}}$	0.8467	-0.561
total	**4.234** (product)	$\mathbf{-0.561}$ (sum)

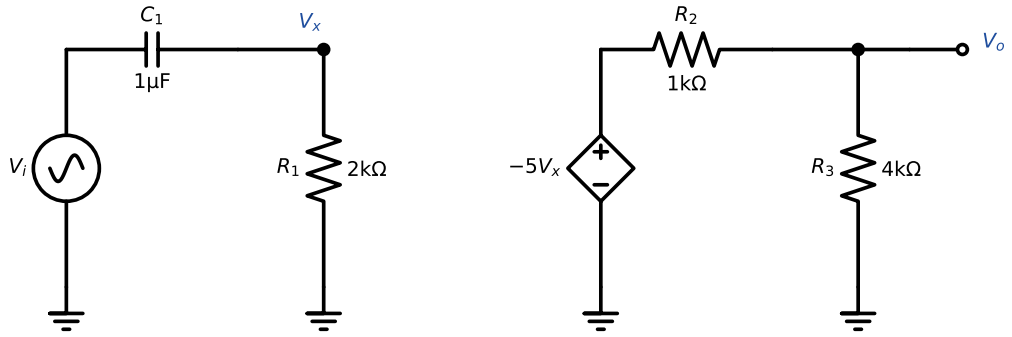
$$|H(j\omega_{in})| = 4.23, \qquad \angle H(j\omega_{in}) = -0.561 \text{ rad}$$
$$v_{o,ac}(t) = 4.23 \times 0.1 \sin(2\pi \cdot 1000\,t - 0.561) = 0.423 \sin(2\pi \cdot 1000\,t - 0.561)$$

$$\boxed{V_o(t) = 0.423 \sin(2\pi \cdot 1000\,t - 0.561)\ \text{V}}$$

Answer

$$H(s) = 5\,\frac{1}{1+\dfrac{s}{2\pi \cdot 1.59\text{k}}} \quad \text{(corners: 1.59 kHz)}$$
$$V_o(t) = 0.423 \sin(2\pi \cdot 1000\,t - 0.561)\ \text{V}$$

Solution 1C.2 **Easy**



Step 1 — Identify the sources

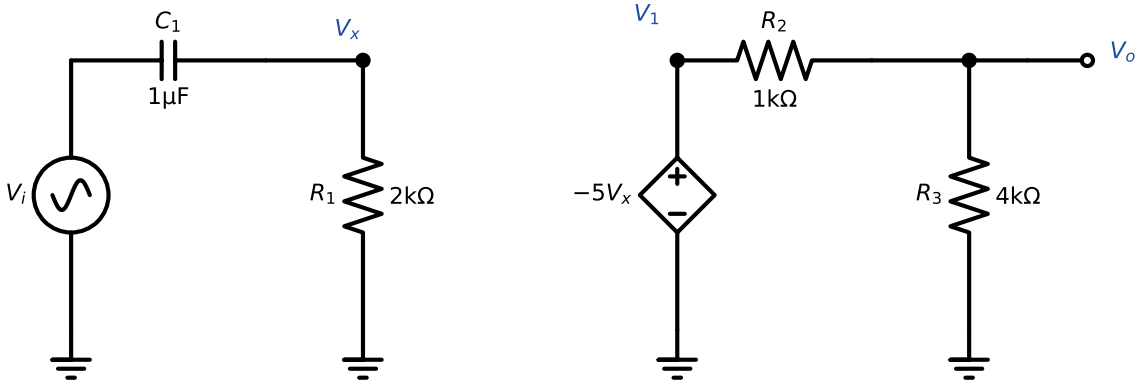
Independent sources: the AC input $V_i(t)$. The diamond-shaped sources are *dependent* sources: they are not inputs, they only copy/scale a voltage that already exists in the circuit, so they stay in the circuit in every step. There is no DC source, so $V_{o,dc} = 0$ and $V_o(t) = v_{o,ac}(t)$: Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — AC circuit

Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL. AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_x \to V_1 \to V_o$. The transfer function is the product of the node-to-node ratios (each one is a simple divider, tool T2, or a dependent source, tool T4):

$$H(s) = \frac{V_o}{V_i} = \frac{V_x}{V_i} \cdot \frac{V_1}{V_x} \cdot \frac{V_o}{V_1}$$

Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_x}{V_i} = \frac{R_1}{\frac{1}{sC_1} + R_1} = \frac{s}{s + 500}$$

$$\frac{V_1}{V_x} = -5 \qquad \text{(ideal VCVS: } V_1 = -5V_x\text{)}$$
$$\frac{V_o}{V_1} = \frac{R_3}{R_2 + R_3} = 0.8$$

Multiply the links and simplify (common factors cancel):

$$H(s) = \frac{-4\,s}{s + 500}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Denominator: $s + 500$; $s + 500 = 0 \Rightarrow s = -500$.

Factored form (leading coefficient in front):

$$H(s) = -4\,\frac{s}{(s + 500)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = -4\,\frac{s}{s + 2\pi \cdot 79.6}, \qquad K = -4$$

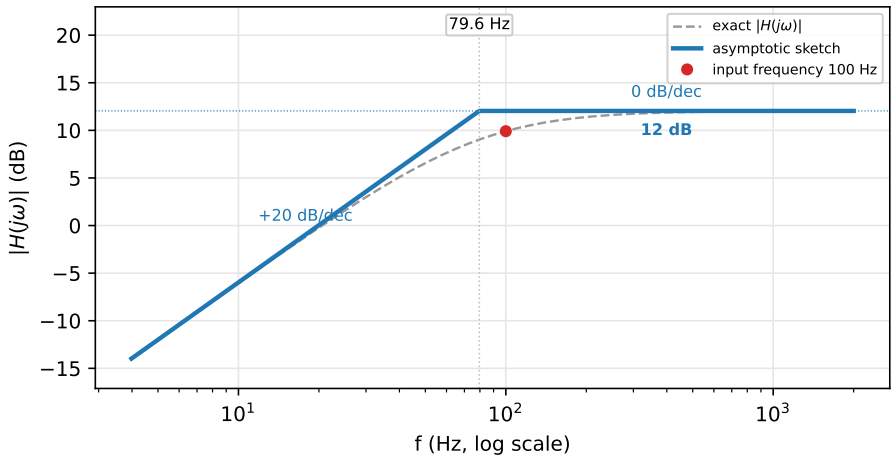
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	500	79.6 Hz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 0.008\,s$, so the plot starts with a slope of +20 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 79.6$ Hz	+20 dB/dec	12 dB at end
$f > 79.6$ Hz	0 dB/dec (flat)	12 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10} 4 = 12$ dB.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.2\sin(2\pi \cdot 100\,t)$, so $\omega_{in} = 2\pi \cdot 100 = 628\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-4	4	-3.14
$\dfrac{s}{s+2\pi\cdot 79.6}$	0.7825	0.672
total	**3.13** (product)	$\mathbf{-2.47}$ (sum)

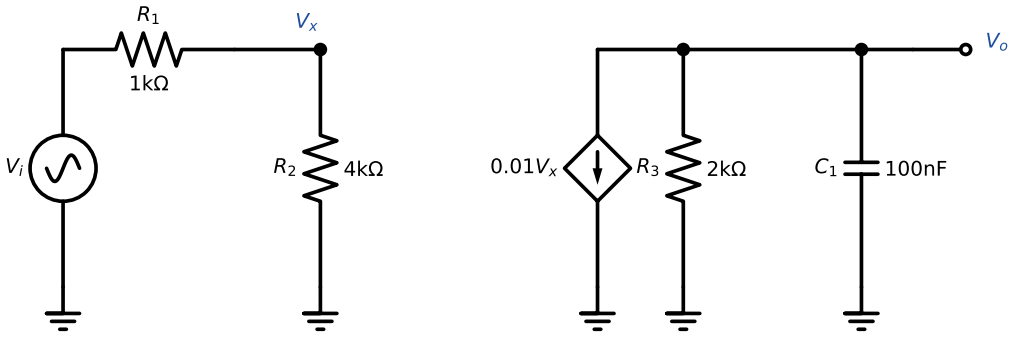
$$|H(j\omega_{in})| = 3.13, \qquad \angle H(j\omega_{in}) = -2.47 \text{ rad}$$
$$v_{o,ac}(t) = 3.13 \times 0.2\,\sin(2\pi\cdot 100\,t - 2.47) = 0.626\,\sin(2\pi\cdot 100\,t - 2.47)$$

$$\boxed{V_o(t) = 0.626\,\sin(2\pi\cdot 100\,t - 2.47)\ \text{V}}$$

Answer

$$H(s) = -4\,\frac{s}{s+2\pi\cdot 79.6} \quad \text{(corners: 79.6 Hz)}$$
$$V_o(t) = 0.626\,\sin(2\pi\cdot 100\,t - 2.47)\ \text{V}$$

Solution 1C.3 **Easy**



Step 1 — Identify the sources

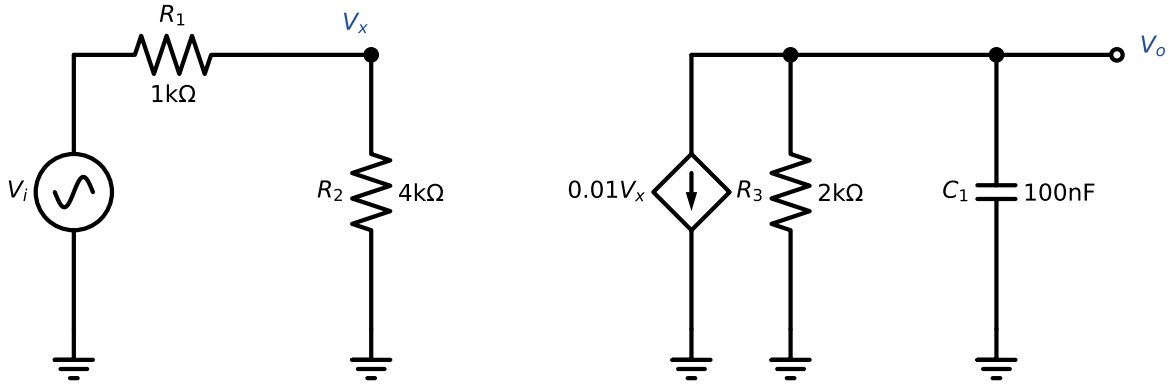
Independent sources: the AC input $V_i(t)$. The diamond-shaped sources are *dependent* sources: they are not inputs, they only copy/scale a voltage that already exists in the circuit, so they stay in the circuit in every step. There is no DC source, so $V_{o,dc} = 0$ and $V_o(t) = v_{o,ac}(t)$: Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — AC circuit

Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL. AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_x \to V_o$. The transfer function is the product of the node-to-node ratios (each one is a simple divider, tool T2, or a dependent source, tool T4):

$$H(s) = \frac{V_o}{V_i} = \frac{V_x}{V_i} \cdot \frac{V_o}{V_x}$$

Impedances seen *looking right* from a node (everything from that node to ground, excluding what is on its left). Compute them from the far right toward the input:

$$Z_o = R_3 \parallel \frac{1}{sC_1} = \frac{10^7}{s + 5000}$$

Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_x}{V_i} = \frac{R_2}{R_1 + R_2} = 0.8$$
$$\frac{V_o}{V_x} = -0.01\, Z_o = \frac{-10^5}{s + 5000}$$

(VCCS: the current $0.01V_x$ flows out of node V_o and through Z_o to ground, so $V_o = -0.01V_x\, Z_o$ by Ohm's law.) Multiply the links and simplify (common factors cancel):

$$H(s) = \frac{-80000}{s + 5000}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Denominator: $s + 5000$; $s + 5000 = 0 \Rightarrow s = -5000$.

Factored form (leading coefficient in front):

$$H(s) = -80000\, \frac{1}{(s + 5000)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = -16\, \frac{1}{1 + \dfrac{s}{2\pi \cdot 796}}, \qquad K = -16$$

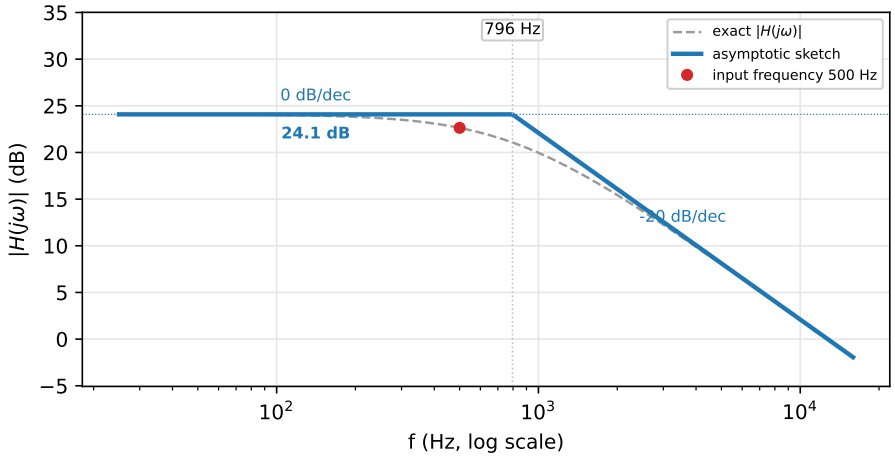
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	5000	796 Hz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 16 = 24.1$ dB (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 796$ Hz	0 dB/dec (flat)	24.1 dB at end
$f > 796$ Hz	−20 dB/dec	24.1 dB at start

The flat (pass-band) level is $20 \log_{10} |K| = 20 \log_{10} 16 = 24.1$ dB.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.05\sin(2\pi \cdot 500\,t)$, so $\omega_{in} = 2\pi \cdot 500 = 3140\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-16	16	-3.14
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 796}}$	0.8467	-0.561
total	**13.55** (product)	$-$**3.7** (sum)

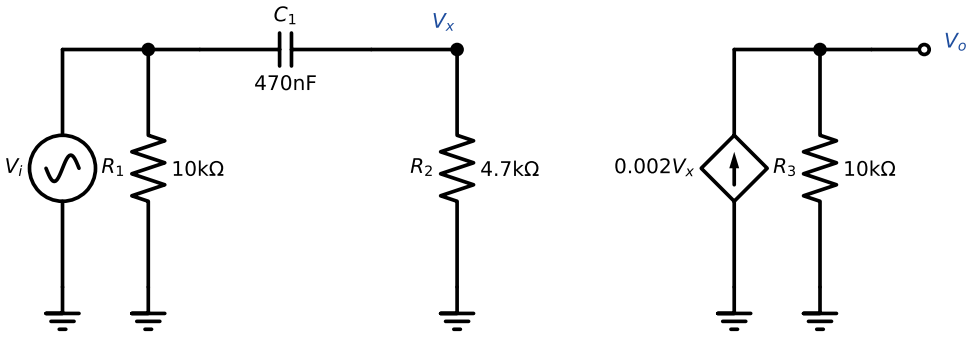
$$|H(j\omega_{in})| = 13.5, \qquad \angle H(j\omega_{in}) = -3.7 \text{ rad}$$
$$v_{o,ac}(t) = 13.5 \times 0.05 \sin(2\pi \cdot 500\,t - 3.7) = 0.677 \sin(2\pi \cdot 500\,t - 3.7)$$

$$\boxed{V_o(t) = 0.677 \sin(2\pi \cdot 500\,t - 3.7)\ \text{V}}$$

Answer

$$H(s) = -16\,\frac{1}{1+\dfrac{s}{2\pi \cdot 796}} \quad \text{(corners: 796 Hz)}$$
$$V_o(t) = 0.677 \sin(2\pi \cdot 500\,t - 3.7)\ \text{V}$$

Solution 1C.4 **Easy**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. The diamond-shaped sources are *dependent* sources: they are not inputs, they only copy/scale a voltage that already exists in the circuit, so they stay in the circuit in every step. There is no DC source, so $V_{o,dc} = 0$ and $V_o(t) = v_{o,ac}(t)$: Step 2 is skipped.

Step 2 — DC circuit

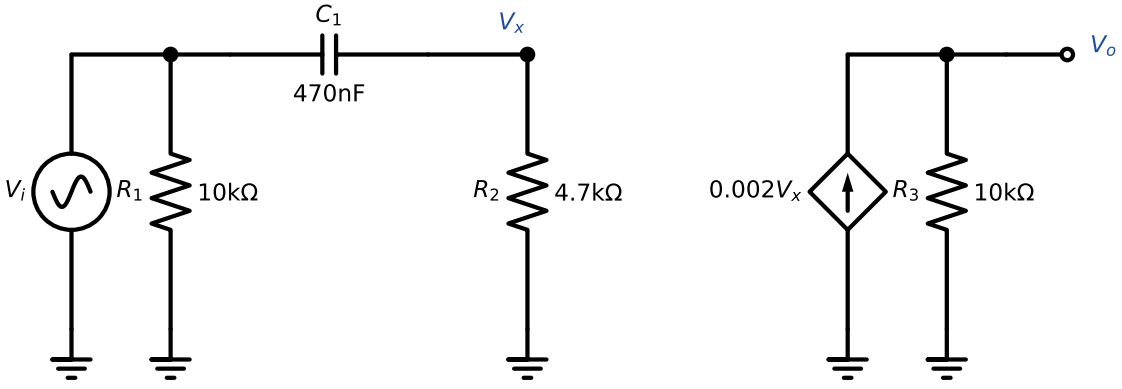
No DC sources: $V_{o,dc} = 0$.

Step 3 — AC circuit

Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL.

- R_1 sits directly across the ideal input source V_i. An ideal voltage source keeps its voltage whatever you connect to it, so R_1 cannot change any node voltage: it drops out of $H(s)$.

AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_x \to V_o$. The transfer function is the product of the node-to-node ratios (each one is a simple divider, tool T2, or a dependent source, tool T4):

$$H(s) = \frac{V_o}{V_i} = \frac{V_x}{V_i} \cdot \frac{V_o}{V_x}$$

Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with

a plain s^n.)

$$\frac{V_x}{V_i} = \frac{R_2}{\frac{1}{sC_1} + R_2} = \frac{s}{s + 453}$$
$$\frac{V_o}{V_x} = 0.002\, R_3 = 20$$

(VCCS: the current $0.002V_x$ flows into node V_o and through R_3 to ground, so $V_o = 0.002V_x\, R_3$ by Ohm's law.) Multiply the links and simplify (common factors cancel):

$$H(s) = \frac{20\, s}{s + 453}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Denominator: $s + 453$; $s + 453 = 0 \Rightarrow s = -453$.

Factored form (leading coefficient in front):

$$H(s) = 20\, \frac{s}{(s + 453)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 20\, \frac{s}{s + 2\pi \cdot 72}, \qquad K = 20$$

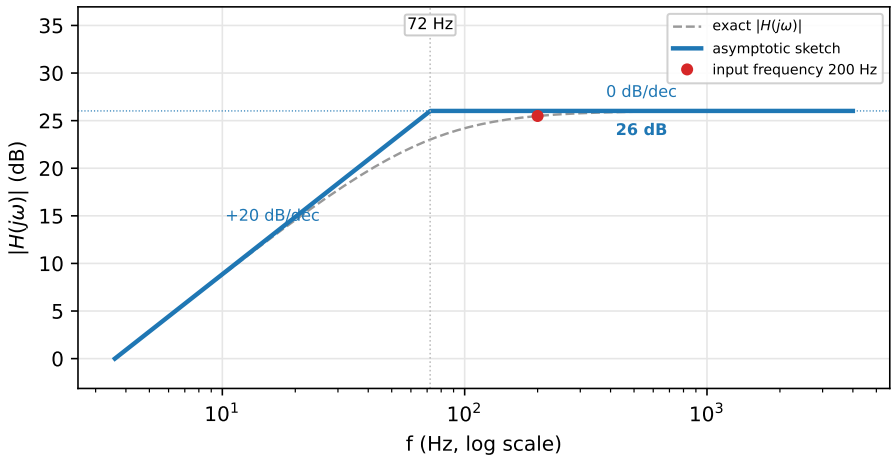
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	453	72 Hz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 0.0442\, s$, so the plot starts with a slope of +20 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 72$ Hz	+20 dB/dec	26 dB at end
$f > 72$ Hz	0 dB/dec (flat)	26 dB at start

The flat (pass-band) level is $20 \log_{10} |K| = 20 \log_{10} 20 = 26$ dB.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.1\sin(2\pi \cdot 200\,t)$, so $\omega_{in} = 2\pi \cdot 200 = 1260\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
20	20	0
$\dfrac{s}{s+2\pi\cdot 72}$	0.9408	0.346
total	**18.82** (product)	**0.346** (sum)

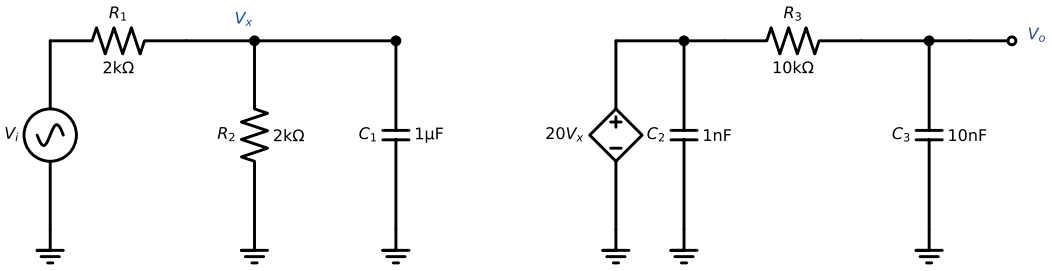
$$|H(j\omega_{in})| = 18.8, \qquad \angle H(j\omega_{in}) = 0.346 \text{ rad}$$
$$v_{o,ac}(t) = 18.8 \times 0.1 \sin(2\pi \cdot 200\,t + 0.346) = 1.88 \sin(2\pi \cdot 200\,t + 0.346)$$

$$\boxed{V_o(t) = 1.88 \sin(2\pi \cdot 200\,t + 0.346)\ \text{V}}$$

Answer

$$H(s) = 20\,\frac{s}{s+2\pi\cdot 72} \quad \text{(corners: 72 Hz)}$$
$$V_o(t) = 1.88 \sin(2\pi \cdot 200\,t + 0.346)\ \text{V}$$

Solution 1C.5 **Easy**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. The diamond-shaped sources are *dependent* sources: they are not inputs, they only copy/scale a voltage that already exists in the circuit, so they stay in the circuit in every step. There is no DC source, so $V_{o,dc} = 0$ and $V_o(t) = v_{o,ac}(t)$: Step 2 is skipped.

Step 2 — DC circuit

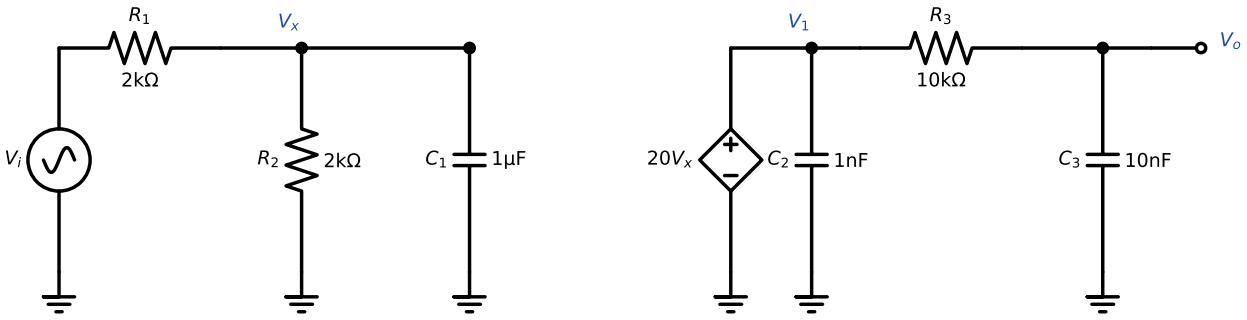
No DC sources: $V_{o,dc} = 0$.

Step 3 — AC circuit

Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL.

- C_2 sits directly across the output of the ideal dependent voltage source. An ideal voltage source keeps its voltage whatever you connect to it, so C_2 cannot change any node voltage: it drops out of $H(s)$.

AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_x \to V_1 \to V_o$. The transfer function is the product of the node-to-node ratios (each one is a simple divider, tool T2, or a dependent source, tool T4):

$$H(s) = \frac{V_o}{V_i} = \frac{V_x}{V_i} \cdot \frac{V_1}{V_x} \cdot \frac{V_o}{V_1}$$

Impedances seen *looking right* from a node (everything from that node to ground, excluding what is on its left). Compute them from the far right toward the input:

$$Z_x = R_2 \parallel \frac{1}{sC_1} = \frac{10^6}{s + 500}$$

Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_x}{V_i} = \frac{Z_x}{R_1 + Z_x} = \frac{500}{s + 1000}$$

$$\frac{V_1}{V_x} = 20 \qquad \text{(ideal VCVS: } V_1 = 20V_x\text{)}$$

$$\frac{V_o}{V_1} = \frac{\frac{1}{sC_3}}{R_3 + \frac{1}{sC_3}} = \frac{10000}{s + 10000}$$

Multiply the links and simplify (common factors cancel):

$$H(s) = \frac{10^8}{s^2 + 11000\,s + 10^7}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Denominator: $s^2 + 11000\,s + 10^7$; quadratic formula:

$$s = \frac{-11000 \pm \sqrt{(11000)^2 - 4(10^7)}}{2} = \frac{-11000 \pm 9000}{2} \Rightarrow s = -1000,\ -10000$$

Factored form (leading coefficient in front):

$$H(s) = 10^8\,\frac{1}{(s + 1000)(s + 10000)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 10\,\frac{1}{1 + \dfrac{s}{2\pi \cdot 159}}\,\frac{1}{1 + \dfrac{s}{2\pi \cdot 1.59\text{k}}}, \qquad K = 10$$

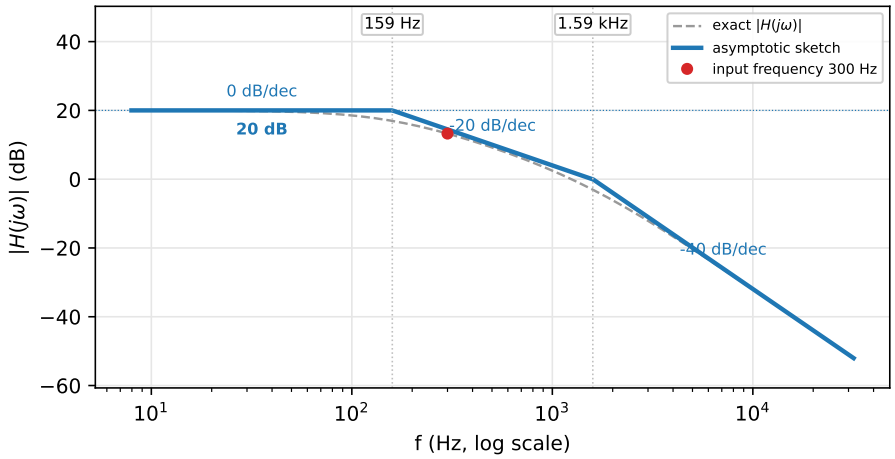
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	1000	159 Hz	−20 dB/dec
pole	10000	1.59 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 10 = 20\,\text{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 159$ Hz	0 dB/dec (flat)	20 dB at end
159 Hz – 1.59 kHz	−20 dB/dec	20 dB at start, 0 dB at end
$f > 1.59$ kHz	−40 dB/dec	0 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10}10 = 20\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.05\sin(2\pi\cdot 300\,t)$, so $\omega_{in} = 2\pi\cdot 300 = 1880\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
10	10	0
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 159}}$	0.4686	-1.08
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 1.59\text{k}}}$	0.9827	-0.186
total	**4.605** (product)	$-$**1.27** (sum)

$$|H(j\omega_{in})| = 4.61, \qquad \angle H(j\omega_{in}) = -1.27\ \text{rad}$$
$$v_{o,ac}(t) = 4.61\times 0.05\,\sin(2\pi\cdot 300\,t - 1.27) = 0.23\,\sin(2\pi\cdot 300\,t - 1.27)$$

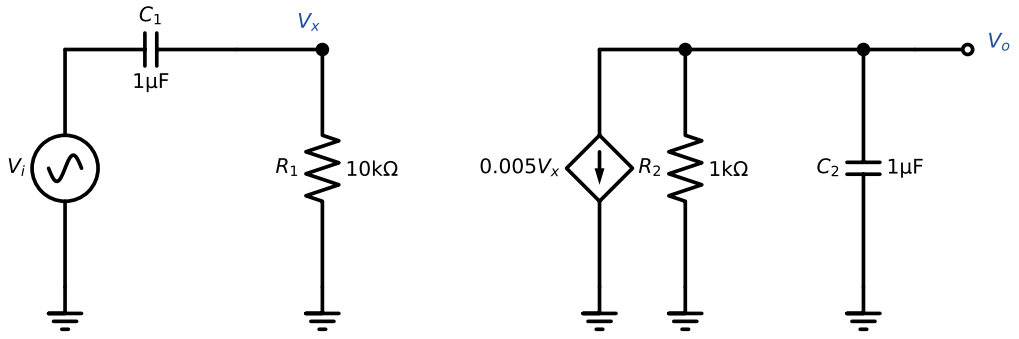
$$\boxed{V_o(t) = 0.23\,\sin(2\pi\cdot 300\,t - 1.27)\ \text{V}}$$

Answer

$$H(s) = 10\,\frac{1}{1+\dfrac{s}{2\pi\cdot 159}}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 1.59\text{k}}} \quad \text{(corners: 159 Hz, 1.59 kHz)}$$

$$V_o(t) = 0.23 \sin\left(2\pi \cdot 300\, t - 1.27\right) \text{ V}$$

Solution 1C.6 **Easy**



Step 1 — Identify the sources

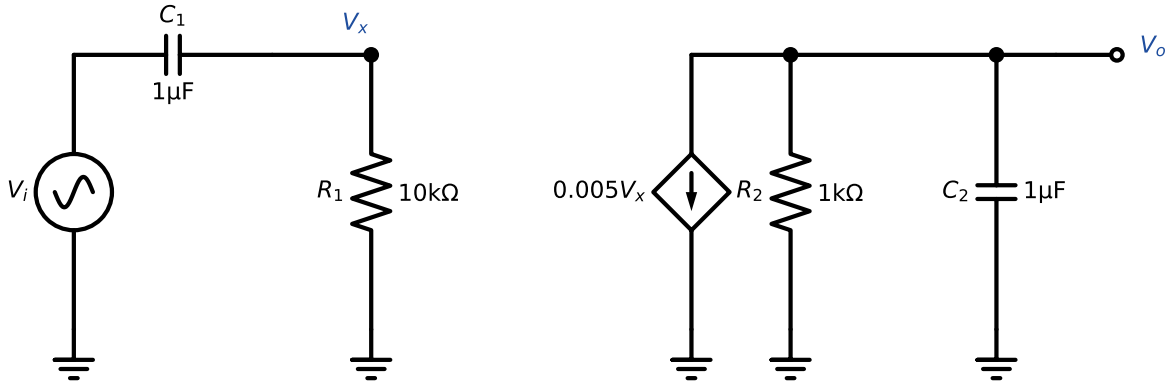
Independent sources: the AC input $V_i(t)$. The diamond-shaped sources are *dependent* sources: they are not inputs, they only copy/scale a voltage that already exists in the circuit, so they stay in the circuit in every step. There is no DC source, so $V_{o,dc} = 0$ and $V_o(t) = v_{o,ac}(t)$: Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — AC circuit

Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL. AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_x \to V_o$. The transfer function is the product of the node-to-node ratios (each one is a simple divider, tool T2, or a dependent source, tool T4):

$$H(s) = \frac{V_o}{V_i} = \frac{V_x}{V_i} \cdot \frac{V_o}{V_x}$$

Impedances seen *looking right* from a node (everything from that node to ground, excluding what is on its left). Compute them from the far right toward the input:

$$Z_o = R_2 \parallel \frac{1}{sC_2} = \frac{10^6}{s + 1000}$$

Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_x}{V_i} = \frac{R_1}{\frac{1}{sC_1} + R_1} = \frac{s}{s + 100}$$
$$\frac{V_o}{V_x} = -0.005\, Z_o = \frac{-5000}{s + 1000}$$

(VCCS: the current $0.005V_x$ flows out of node V_o and through Z_o to ground, so $V_o = -0.005V_x\, Z_o$ by Ohm's law.) Multiply the links and simplify (common factors cancel):

$$H(s) = \frac{-5000\, s}{s^2 + 1100\, s + 10^5}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Denominator: $s^2 + 1100\, s + 10^5$; quadratic formula:

$$s = \frac{-1100 \pm \sqrt{(1100)^2 - 4(10^5)}}{2} = \frac{-1100 \pm 900}{2} \Rightarrow s = -100,\ -1000$$

Factored form (leading coefficient in front):

$$H(s) = -5000\, \frac{s}{(s + 100)(s + 1000)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = -5\, \frac{s}{s + 2\pi \cdot 15.9}\, \frac{1}{1 + \dfrac{s}{2\pi \cdot 159}}, \qquad K = -5$$

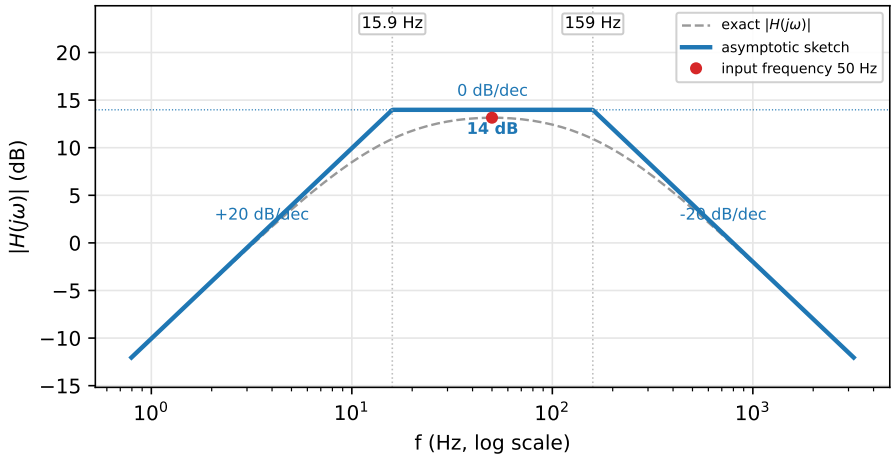
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	100	15.9 Hz	−20 dB/dec
pole	1000	159 Hz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 0.05\, s$, so the plot starts with a slope of +20 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 15.9$ Hz	+20 dB/dec	14 dB at end
15.9 Hz – 159 Hz	0 dB/dec (flat)	14 dB at start, 14 dB at end
$f > 159$ Hz	−20 dB/dec	14 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10}5 = 14\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.2\sin(2\pi\cdot 50\,t)$, so $\omega_{in} = 2\pi\cdot 50 = 314\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-5	5	-3.14
$\dfrac{s}{s+2\pi\cdot 15.9}$	0.9529	0.308
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 159}}$	0.954	-0.304
total	**4.545** (product)	$-$**3.14** (sum)

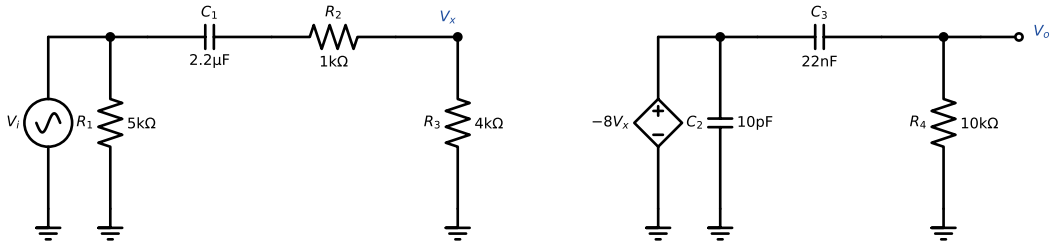
$$|H(j\omega_{in})| = 4.55, \qquad \angle H(j\omega_{in}) = -3.14 \text{ rad}$$
$$v_{o,ac}(t) = 4.55\times 0.2\,\sin(2\pi\cdot 50\,t - 3.14) = 0.909\,\sin(2\pi\cdot 50\,t - 3.14)$$

$$\boxed{V_o(t) = 0.909\,\sin(2\pi\cdot 50\,t - 3.14)\ \text{V}}$$

Answer

$$H(s) = -5\,\frac{s}{s+2\pi\cdot 15.9}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 159}} \quad \text{(corners: 15.9 Hz, 159 Hz)}$$
$$V_o(t) = 0.909\,\sin(2\pi\cdot 50\,t - 3.14)\ \text{V}$$

Solution 1C.7 **Medium**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. The diamond-shaped sources are *dependent* sources: they are not inputs, they only copy/scale a voltage that already exists in the circuit, so they stay in the circuit in every step. There is no DC source, so $V_{o,dc} = 0$ and $V_o(t) = v_{o,ac}(t)$: Step 2 is skipped.

Step 2 — DC circuit

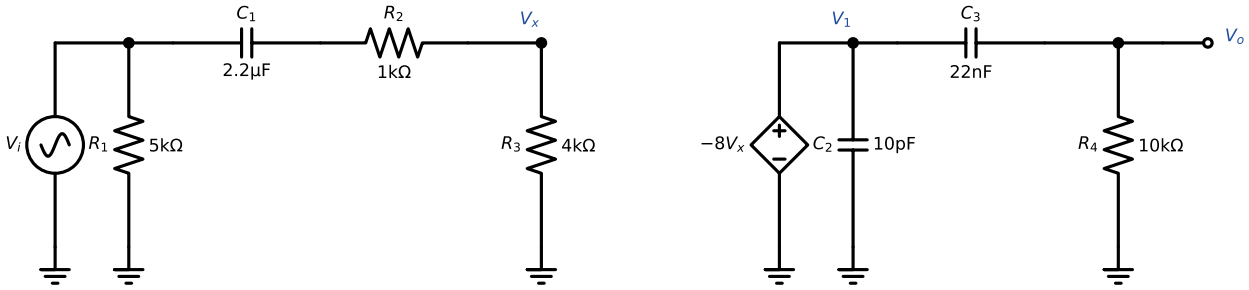
No DC sources: $V_{o,dc} = 0$.

Step 3 — AC circuit

Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL.

- R_1 sits directly across the ideal input source V_i. An ideal voltage source keeps its voltage whatever you connect to it, so R_1 cannot change any node voltage: it drops out of $H(s)$.
- C_2 sits directly across the output of the ideal dependent voltage source. An ideal voltage source keeps its voltage whatever you connect to it, so C_2 cannot change any node voltage: it drops out of $H(s)$.

AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_x \to V_1 \to V_o$. The transfer function is the product of the node-to-node ratios (each one is a simple divider, tool T2, or a dependent source, tool T4):

$$H(s) = \frac{V_o}{V_i} = \frac{V_x}{V_i} \cdot \frac{V_1}{V_x} \cdot \frac{V_o}{V_1}$$

Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_x}{V_i} = \frac{R_3}{\frac{1}{sC_1} + R_2 + R_3} = \frac{0.8\,s}{s + 90.9}$$

$$\frac{V_1}{V_x} = -8 \qquad \text{(ideal VCVS: } V_1 = -8V_x\text{)}$$
$$\frac{V_o}{V_1} = \frac{R_4}{\frac{1}{sC_3} + R_4} = \frac{s}{s + 4550}$$

Multiply the links and simplify (common factors cancel):

$$H(s) = \frac{-6.4\, s^2}{s^2 + 4640\, s + 4.13 \times 10^5}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s^2 \Rightarrow$ 2 zeros at $s = 0$.
- Denominator: $s^2 + 4640\, s + 4.13 \times 10^5$; quadratic formula:

$$s = \frac{-4640 \pm \sqrt{(4640)^2 - 4(4.13 \times 10^5)}}{2} = \frac{-4640 \pm 4450}{2} \Rightarrow s = -90.9,\ -4550$$

Factored form (leading coefficient in front):

$$H(s) = -6.4\, \frac{s^2}{(s + 90.9)(s + 4550)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = -6.4\, \frac{s}{s + 2\pi \cdot 14.5}\, \frac{s}{s + 2\pi \cdot 723}, \qquad K = -6.4$$

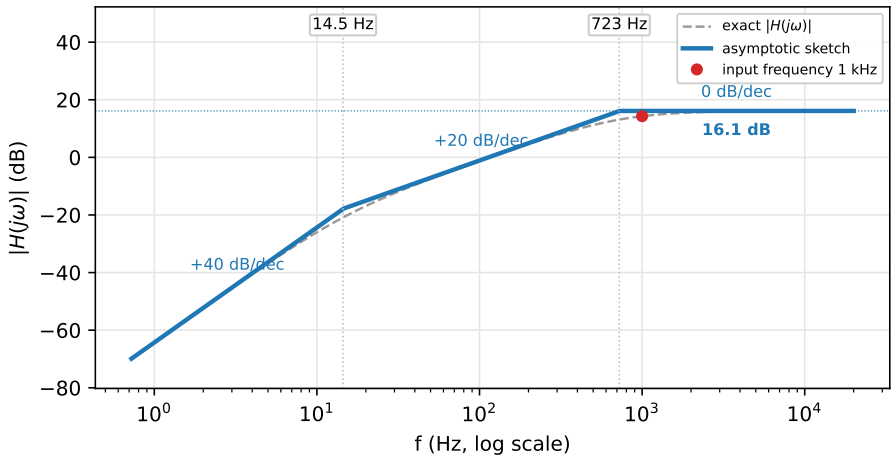
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	90.9	14.5 Hz	−20 dB/dec
pole	4550	723 Hz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 1.55 \times 10^{-5}\, s^2$, so the plot starts with a slope of +40 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f <$ 14.5 Hz	+40 dB/dec	-17.9 dB at end
14.5 Hz – 723 Hz	+20 dB/dec	-17.9 dB at start, 16.1 dB at end
$f >$ 723 Hz	0 dB/dec (flat)	16.1 dB at start

The flat (pass-band) level is $20 \log_{10} |K| = 20 \log_{10} 6.4 = 16.1$ dB.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.1\sin(2\pi \cdot 1000\,t)$, so $\omega_{in} = 2\pi \cdot 1000 = 6280$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-6.4	6.4	-3.14
$\dfrac{s}{s+2\pi\cdot 14.5}$	0.9999	0.0145
$\dfrac{s}{s+2\pi\cdot 723}$	0.8102	0.626
total	**5.185** (product)	$-$**2.5** (sum)

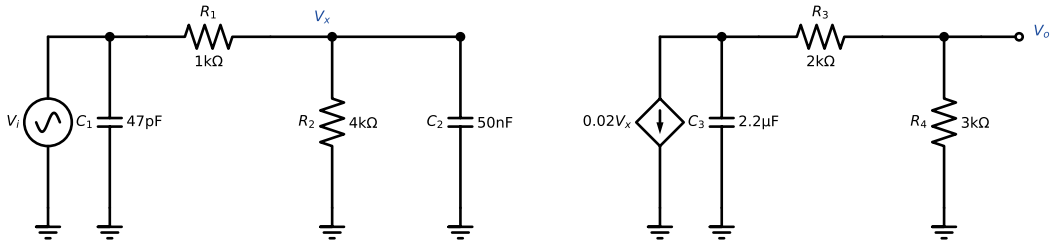
$$|H(j\omega_{in})| = 5.18, \qquad \angle H(j\omega_{in}) = -2.5 \text{ rad}$$
$$v_{o,ac}(t) = 5.18 \times 0.1 \sin(2\pi \cdot 1000\,t - 2.5) = 0.518 \sin(2\pi \cdot 1000\,t - 2.5)$$

$$\boxed{V_o(t) = 0.518 \sin(2\pi \cdot 1000\,t - 2.5) \text{ V}}$$

Answer

$$H(s) = -6.4\,\frac{s}{s+2\pi\cdot 14.5}\,\frac{s}{s+2\pi\cdot 723} \quad \text{(corners: 14.5 Hz, 723 Hz)}$$
$$V_o(t) = 0.518 \sin(2\pi \cdot 1000\,t - 2.5) \text{ V}$$

Solution 1C.8 **Medium**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. The diamond-shaped sources are *dependent* sources: they are not inputs, they only copy/scale a voltage that already exists in the circuit, so they stay in the circuit in every step. There is no DC source, so $V_{o,dc} = 0$ and $V_o(t) = v_{o,ac}(t)$: Step 2 is skipped.

Step 2 — DC circuit

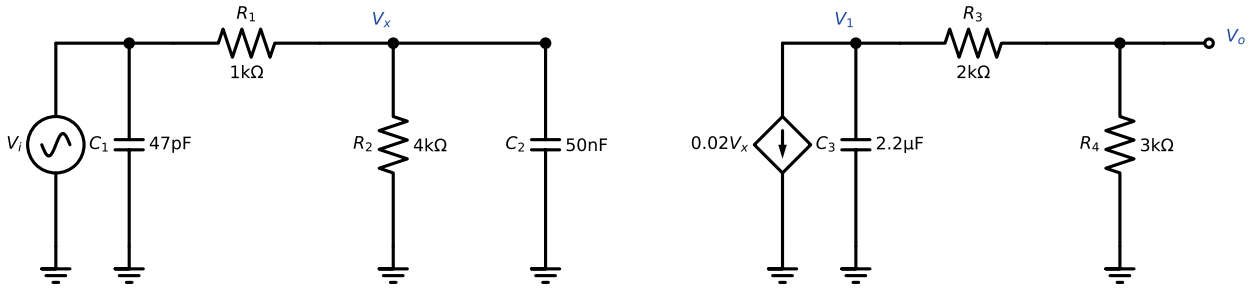
No DC sources: $V_{o,dc} = 0$.

Step 3 — AC circuit

Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL.

- C_1 sits directly across the ideal input source V_i. An ideal voltage source keeps its voltage whatever you connect to it, so C_1 cannot change any node voltage: it drops out of $H(s)$.

AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_x \to V_1 \to V_o$. The transfer function is the product of the node-to-node ratios (each one is a simple divider, tool T2, or a dependent source, tool T4):

$$H(s) = \frac{V_o}{V_i} = \frac{V_x}{V_i} \cdot \frac{V_1}{V_x} \cdot \frac{V_o}{V_1}$$

Impedances seen *looking right* from a node (everything from that node to ground, excluding what is on its left). Compute them from the far right toward the input:

$$Z_x = R_2 \parallel \frac{1}{sC_2} = \frac{2 \times 10^7}{s + 5000}$$

$$Z_1 = \frac{1}{sC_3} \parallel (R_3 + R_4) = \frac{4.55 \times 10^5}{s + 90.9}$$

Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_x}{V_i} = \frac{Z_x}{R_1 + Z_x} = \frac{20000}{s + 25000}$$
$$\frac{V_1}{V_x} = -0.02\, Z_1 = \frac{-9090}{s + 90.9}$$
$$\frac{V_o}{V_1} = \frac{R_4}{R_3 + R_4} = 0.6$$

(VCCS: the current $0.02V_x$ flows out of node V_1 and through Z_1 to ground, so $V_1 = -0.02V_x\, Z_1$ by Ohm's law.) Multiply the links and simplify (common factors cancel):

$$H(s) = \frac{-1.09 \times 10^8}{s^2 + 25100\, s + 2.27 \times 10^6}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Denominator: $s^2 + 25100\, s + 2.27 \times 10^6$; quadratic formula:

$$s = \frac{-25100 \pm \sqrt{(25100)^2 - 4(2.27 \times 10^6)}}{2} = \frac{-25100 \pm 24900}{2} \Rightarrow s = -90.9,\ -25000$$

Factored form (leading coefficient in front):

$$H(s) = -1.09 \times 10^8\, \frac{1}{(s + 90.9)(s + 25000)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = -48\, \frac{1}{1 + \dfrac{s}{2\pi \cdot 14.5}}\, \frac{1}{1 + \dfrac{s}{2\pi \cdot 3.98\text{k}}}, \qquad K = -48$$

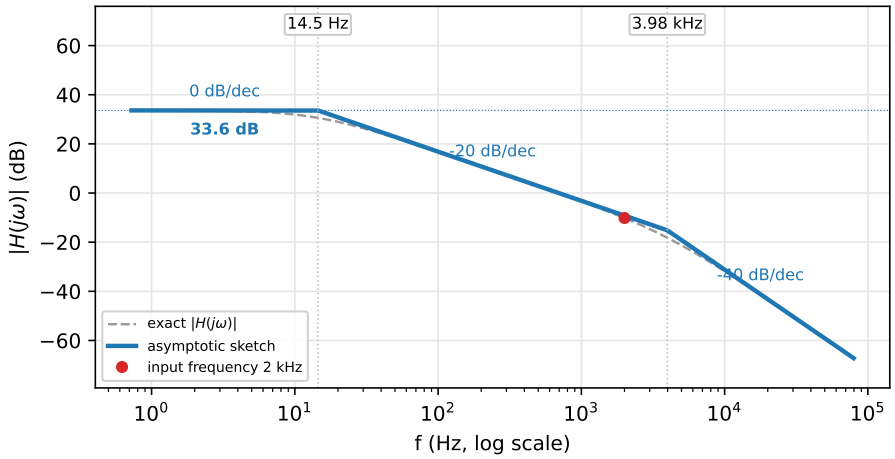
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	90.9	14.5 Hz	−20 dB/dec
pole	25000	3.98 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 48 = 33.6$ dB (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 14.5$ Hz	0 dB/dec (flat)	33.6 dB at end
14.5 Hz – 3.98 kHz	−20 dB/dec	33.6 dB at start, -15.2 dB at end
$f > 3.98$ kHz	−40 dB/dec	-15.2 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10}48 = 33.6\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.05\sin(2\pi\cdot 2000\,t)$, so $\omega_{in} = 2\pi\cdot 2000 = 12600\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-48	48	-3.14
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 14.5}}$	0.007234	-1.56
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 3.98\text{k}}}$	0.8935	-0.466
total	**0.3102** (product)	$\mathbf{-5.17}$ (sum)

$$|H(j\omega_{in})| = 0.31, \qquad \angle H(j\omega_{in}) = -5.17 \text{ rad}$$
$$v_{o,ac}(t) = 0.31\times 0.05\,\sin(2\pi\cdot 2000\,t - 5.17) = 0.0155\,\sin(2\pi\cdot 2000\,t - 5.17)$$

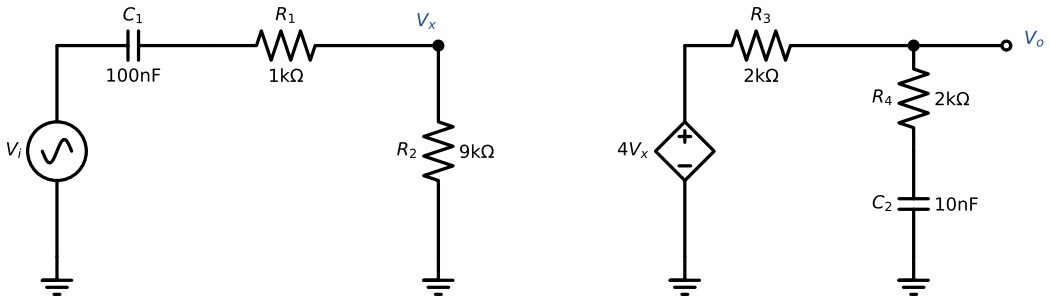
$$\boxed{V_o(t) = 0.0155\,\sin(2\pi\cdot 2000\,t - 5.17)\ \text{V}}$$

Answer

$$H(s) = -48\,\frac{1}{1+\dfrac{s}{2\pi\cdot 14.5}}\,\frac{1}{1+\dfrac{s}{2\pi\cdot 3.98\text{k}}} \quad \text{(corners: 14.5 Hz, 3.98 kHz)}$$

$$V_o(t) = 0.0155 \sin\left(2\pi \cdot 2000\, t - 5.17\right) \text{ V}$$

Solution 1C.9 **Medium**



Step 1 — Identify the sources

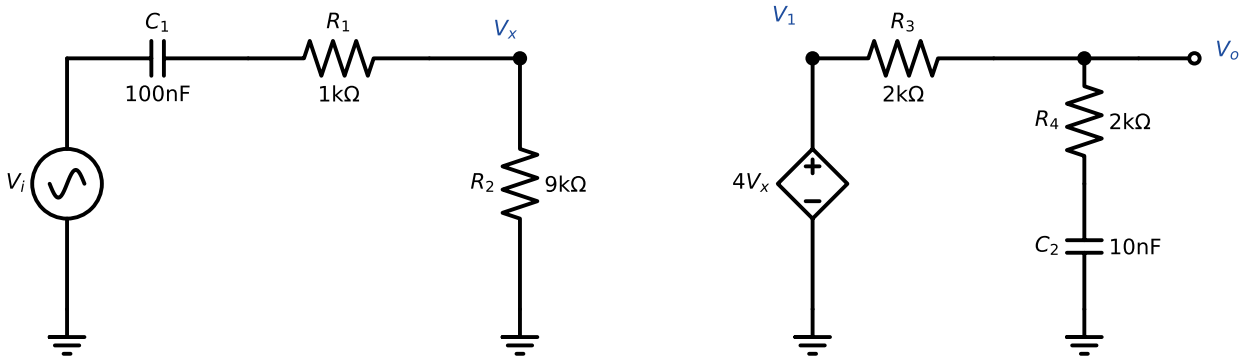
Independent sources: the AC input $V_i(t)$. The diamond-shaped sources are *dependent* sources: they are not inputs, they only copy/scale a voltage that already exists in the circuit, so they stay in the circuit in every step. There is no DC source, so $V_{o,dc} = 0$ and $V_o(t) = v_{o,ac}(t)$: Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — AC circuit

Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL. AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_x \to V_1 \to V_o$. The transfer function is the product of the node-to-node ratios (each one is a simple divider, tool T2, or a dependent source, tool T4):

$$H(s) = \frac{V_o}{V_i} = \frac{V_x}{V_i} \cdot \frac{V_1}{V_x} \cdot \frac{V_o}{V_1}$$

Impedances seen *looking right* from a node (everything from that node to ground, excluding what is on its left). Compute them from the far right toward the input:

$$Z_o = R_4 + \frac{1}{sC_2} = \frac{2000\,s + 10^8}{s}$$

Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with

a plain s^n.)

$$\frac{V_x}{V_i} = \frac{R_2}{\frac{1}{sC_1} + R_1 + R_2} = \frac{0.9\,s}{s + 1000}$$
$$\frac{V_1}{V_x} = 4 \qquad \text{(ideal VCVS: } V_1 = 4V_x\text{)}$$
$$\frac{V_o}{V_1} = \frac{Z_o}{R_3 + Z_o} = \frac{0.5\,s + 25000}{s + 25000}$$

Multiply the links and simplify (common factors cancel):

$$H(s) = \frac{1.8\,s^2 + 90000\,s}{s^2 + 26000\,s + 2.5 \times 10^7}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Numerator (after taking out the s factor): $1.8\,s + 90000$; divide by the leading coefficient 1.8 (this does not move the roots): $s + 50000$; $s + 50000 = 0 \Rightarrow s = -50000$.
- Denominator: $s^2 + 26000\,s + 2.5 \times 10^7$; quadratic formula:

$$s = \frac{-26000 \pm \sqrt{(26000)^2 - 4(2.5 \times 10^7)}}{2} = \frac{-26000 \pm 24000}{2} \Rightarrow s = -1000,\ -25000$$

Factored form (leading coefficient in front):

$$H(s) = 1.8\,\frac{s(s + 50000)}{(s + 1000)(s + 25000)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

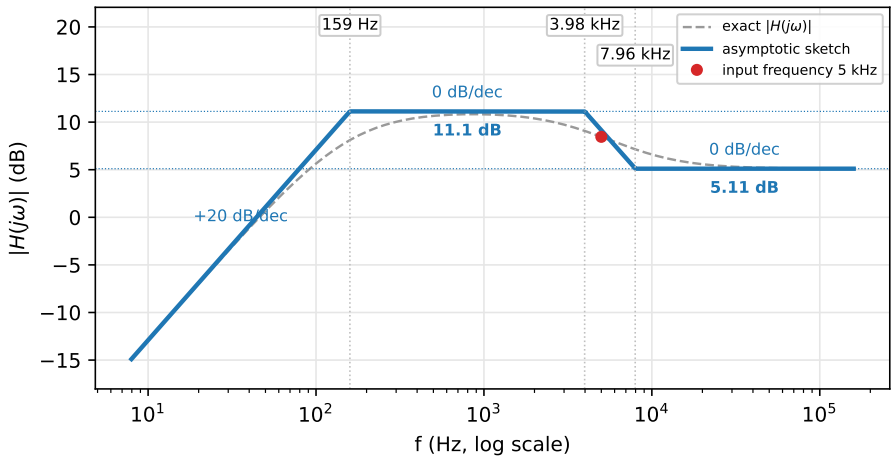
$$H(s) = 3.6\,\frac{s}{s + 2\pi \cdot 159}\,\frac{1}{1 + \frac{s}{2\pi \cdot 3.98\text{k}}}\left(1 + \frac{s}{2\pi \cdot 7.96\text{k}}\right), \qquad K = 3.6$$

type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	1000	159 Hz	−20 dB/dec
pole	25000	3.98 kHz	−20 dB/dec
zero	50000	7.96 kHz	+20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 0.0036\,s$, so the plot starts with a slope of +20 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 159$ Hz	+20 dB/dec	11.1 dB at end
159 Hz – 3.98 kHz	0 dB/dec (flat)	11.1 dB at start, 11.1 dB at end
3.98 kHz – 7.96 kHz	−20 dB/dec	11.1 dB at start, 5.11 dB at end
$f > 7.96$ kHz	0 dB/dec (flat)	5.11 dB at start



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.2\sin(2\pi \cdot 5000\,t)$, so $\omega_{in} = 2\pi \cdot 5000 = 31400$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
3.6	3.6	0
$\dfrac{s}{s+2\pi\cdot 159}$	0.9995	0.0318
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 3.98\text{k}}}$	0.6227	−0.899
$\left(1+\dfrac{s}{2\pi\cdot 7.96\text{k}}\right)$	1.181	0.561
total	**2.646** (product)	**−0.306** (sum)

$$|H(j\omega_{in})| = 2.65, \qquad \angle H(j\omega_{in}) = -0.306 \text{ rad}$$
$$v_{o,ac}(t) = 2.65 \times 0.2\,\sin(2\pi\cdot 5000\,t - 0.306) = 0.529\,\sin(2\pi\cdot 5000\,t - 0.306)$$

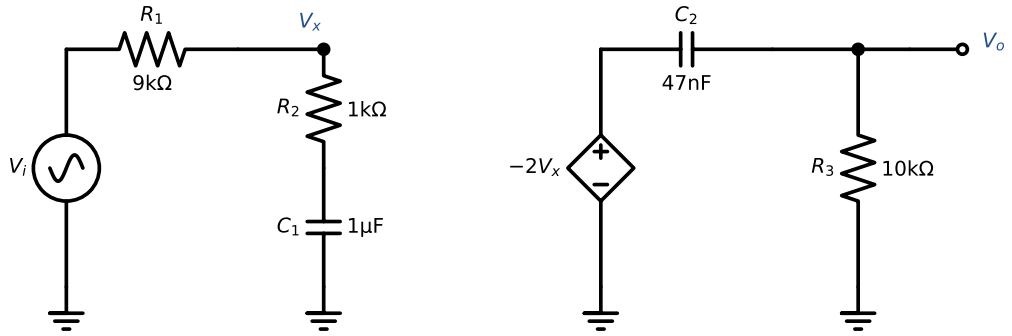
$$\boxed{V_o(t) = 0.529\ \sin\left(2\pi \cdot 5000\, t - 0.306\right)\ \text{V}}$$

Answer

$$H(s) = 3.6\,\frac{s}{s + 2\pi \cdot 159}\,\frac{1}{1 + \dfrac{s}{2\pi \cdot 3.98\text{k}}}\left(1 + \frac{s}{2\pi \cdot 7.96\text{k}}\right) \quad \text{(corners: 159 Hz, 3.98 kHz, 7.96 kHz)}$$

$$V_o(t) = 0.529\ \sin\left(2\pi \cdot 5000\, t - 0.306\right)\ \text{V}$$

Solution 1C.10 **Medium**



Step 1 — Identify the sources

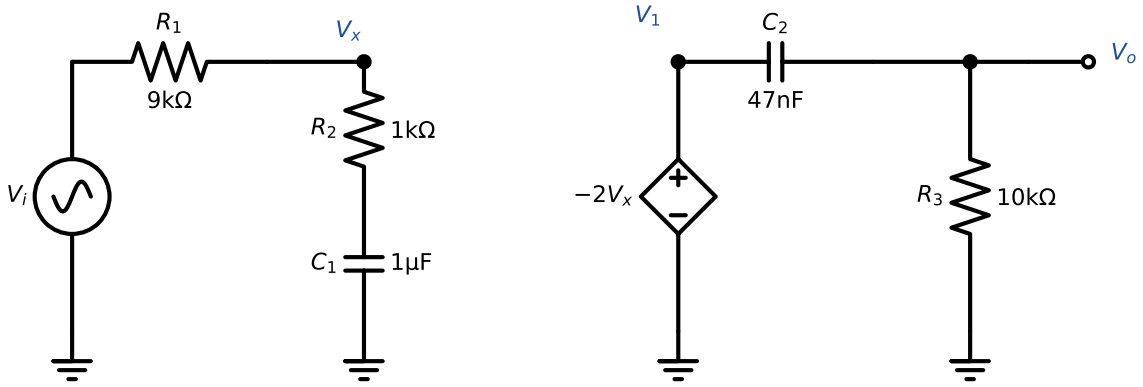
Independent sources: the AC input $V_i(t)$. The diamond-shaped sources are *dependent* sources: they are not inputs, they only copy/scale a voltage that already exists in the circuit, so they stay in the circuit in every step. There is no DC source, so $V_{o,dc} = 0$ and $V_o(t) = v_{o,ac}(t)$: Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — AC circuit

Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL. AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_x \to V_1 \to V_o$. The transfer function is the product of the node-to-node ratios (each one is a simple divider, tool T2, or a dependent source, tool T4):

$$H(s) = \frac{V_o}{V_i} = \frac{V_x}{V_i} \cdot \frac{V_1}{V_x} \cdot \frac{V_o}{V_1}$$

Impedances seen *looking right* from a node (everything from that node to ground, excluding what is on its left). Compute them from the far right toward the input:

$$Z_x = R_2 + \frac{1}{sC_1} = \frac{1000\,s + 10^6}{s}$$

Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_x}{V_i} = \frac{Z_x}{R_1 + Z_x} = \frac{0.1\,s + 100}{s + 100}$$
$$\frac{V_1}{V_x} = -2 \qquad \text{(ideal VCVS: } V_1 = -2V_x\text{)}$$
$$\frac{V_o}{V_1} = \frac{R_3}{\frac{1}{sC_2} + R_3} = \frac{s}{s + 2130}$$

Multiply the links and simplify (common factors cancel):

$$H(s) = \frac{-0.2\,s^2 - 200\,s}{s^2 + 2230\,s + 2.13 \times 10^5}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Numerator (after taking out the s factor): $-0.2\,s - 200$; divide by the leading coefficient -0.2 (this does not move the roots): $s + 1000$; $s + 1000 = 0 \Rightarrow s = -1000$.
- Denominator: $s^2 + 2230\,s + 2.13 \times 10^5$; quadratic formula:

$$s = \frac{-2230 \pm \sqrt{(2230)^2 - 4(2.13 \times 10^5)}}{2} = \frac{-2230 \pm 2030}{2} \Rightarrow s = -100,\ -2130$$

Factored form (leading coefficient in front):

$$H(s) = -0.2\,\frac{s(s + 1000)}{(s + 100)(s + 2130)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

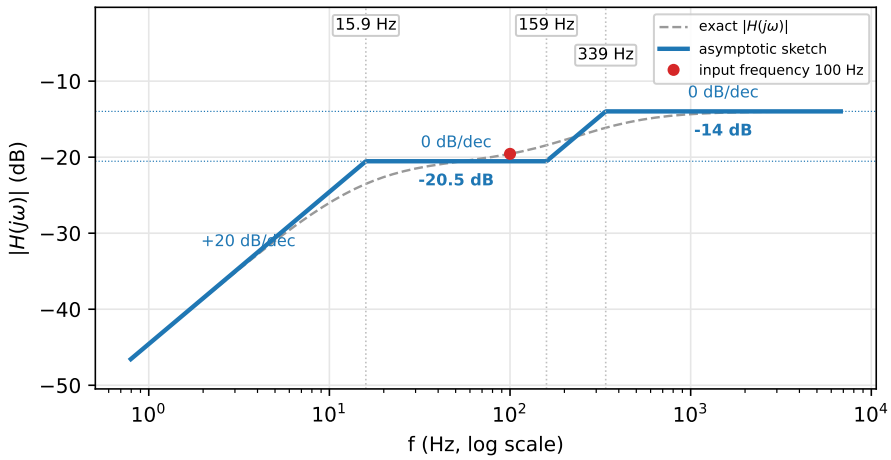
$$H(s) = -0.094\,\frac{s}{s + 2\pi \cdot 15.9}\left(1 + \frac{s}{2\pi \cdot 159}\right)\frac{1}{1 + \frac{s}{2\pi \cdot 339}}, \qquad K = -0.094$$

type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	100	15.9 Hz	−20 dB/dec
zero	1000	159 Hz	+20 dB/dec
pole	2130	339 Hz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 9.4 \times 10^{-4}\,s$, so the plot starts with a slope of +20 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 15.9$ Hz	+20 dB/dec	-20.5 dB at end
15.9 Hz – 159 Hz	0 dB/dec (flat)	-20.5 dB at start, -20.5 dB at end
159 Hz – 339 Hz	+20 dB/dec	-20.5 dB at start, -14 dB at end
$f > 339$ Hz	0 dB/dec (flat)	-14 dB at start



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.5\sin(2\pi \cdot 100\,t)$, so $\omega_{in} = 2\pi \cdot 100 = 628$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-0.094	0.094	-3.14
$\dfrac{s}{s+2\pi\cdot 15.9}$	0.9876	0.158
$\left(1+\dfrac{s}{2\pi\cdot 159}\right)$	1.181	0.561
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 339}}$	0.9591	-0.287
total	**0.1051** (product)	$-$**2.71** (sum)

$$|H(j\omega_{in})| = 0.105, \qquad \angle H(j\omega_{in}) = -2.71 \text{ rad}$$
$$v_{o,ac}(t) = 0.105 \times 0.5\,\sin(2\pi\cdot 100\,t - 2.71) = 0.0526\,\sin(2\pi\cdot 100\,t - 2.71)$$

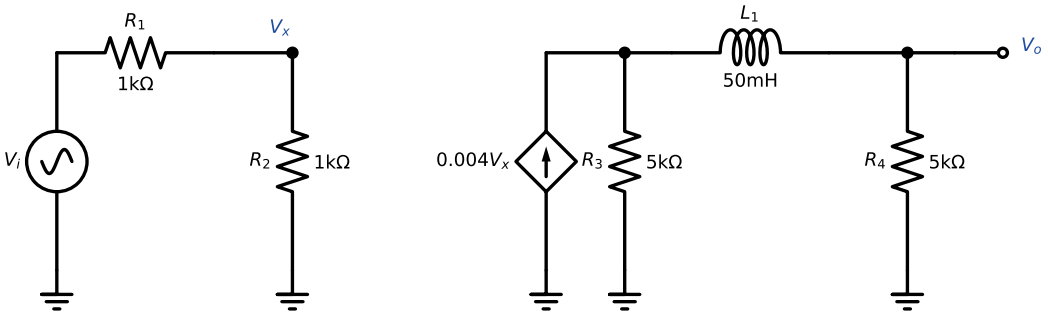
$$\boxed{V_o(t) = 0.0526\,\sin\left(2\pi \cdot 100\,t - 2.71\right)\ \text{V}}$$

Answer

$$H(s) = -0.094\,\frac{s}{s + 2\pi \cdot 15.9}\left(1 + \frac{s}{2\pi \cdot 159}\right)\frac{1}{1 + \dfrac{s}{2\pi \cdot 339}} \quad \text{(corners: 15.9 Hz, 159 Hz, 339 Hz)}$$

$$V_o(t) = 0.0526\,\sin\left(2\pi \cdot 100\,t - 2.71\right)\ \text{V}$$

Solution 1C.11 **Medium**



Step 1 — Identify the sources

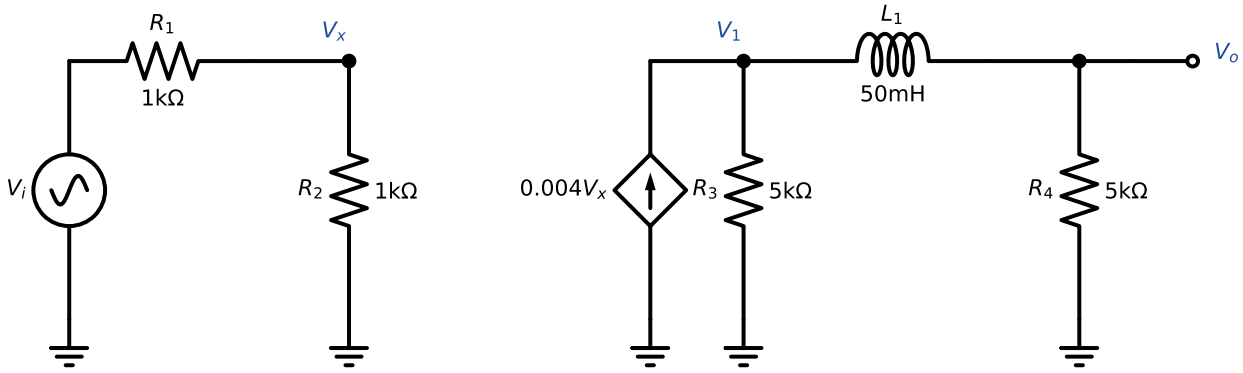
Independent sources: the AC input $V_i(t)$. The diamond-shaped sources are *dependent* sources: they are not inputs, they only copy/scale a voltage that already exists in the circuit, so they stay in the circuit in every step. There is no DC source, so $V_{o,dc} = 0$ and $V_o(t) = v_{o,ac}(t)$: Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — AC circuit

Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL. AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_x \to V_1 \to V_o$. The transfer function is the product of the node-to-node ratios (each one is a simple divider, tool T2, or a dependent source, tool T4):

$$H(s) = \frac{V_o}{V_i} = \frac{V_x}{V_i} \cdot \frac{V_1}{V_x} \cdot \frac{V_o}{V_1}$$

Impedances seen *looking right* from a node (everything from that node to ground, excluding what is on its left). Compute them from the far right toward the input:

$$Z_1 = R_3 \parallel (sL_1 + R_4) = \frac{5000\, s + 5 \times 10^8}{s + 2 \times 10^5}$$

Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with

a plain s^n.)

$$\frac{V_x}{V_i} = \frac{R_2}{R_1 + R_2} = 0.5$$
$$\frac{V_1}{V_x} = 0.004\, Z_1 = \frac{20\, s + 2 \times 10^6}{s + 2 \times 10^5}$$
$$\frac{V_o}{V_1} = \frac{R_4}{sL_1 + R_4} = \frac{10^5}{s + 10^5}$$

(VCCS: the current $0.004V_x$ flows into node V_1 and through Z_1 to ground, so $V_1 = 0.004V_x\, Z_1$ by Ohm's law.) Multiply the links and simplify (common factors cancel):

$$H(s) = \frac{10^6}{s + 2 \times 10^5}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Denominator: $s + 2 \times 10^5$; $s + 2 \times 10^5 = 0 \Rightarrow s = -2 \times 10^5$.

Factored form (leading coefficient in front):

$$H(s) = 10^6\, \frac{1}{(s + 2 \times 10^5)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 5\, \frac{1}{1 + \dfrac{s}{2\pi \cdot 31.8\text{k}}}, \qquad K = 5$$

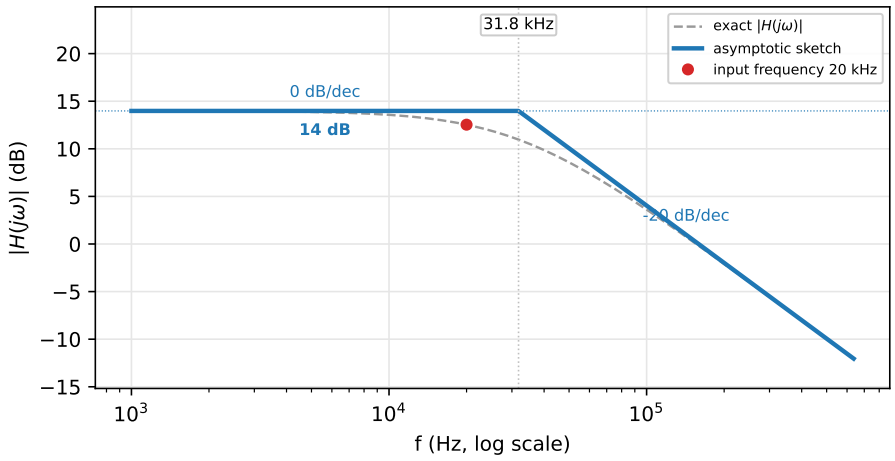
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	2×10^5	31.8 kHz	-20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 5 = 14\,\text{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 31.8$ kHz	0 dB/dec (flat)	14 dB at end
$f > 31.8$ kHz	-20 dB/dec	14 dB at start

The flat (pass-band) level is $20 \log_{10} |K| = 20 \log_{10} 5 = 14\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.2\sin(2\pi \cdot 20000\,t)$, so $\omega_{in} = 2\pi{\cdot}20000 = 1.26{\times}10^5$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
5	5	0
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 31.8\text{k}}}$	0.8467	-0.561
total	**4.234** (product)	**−0.561** (sum)

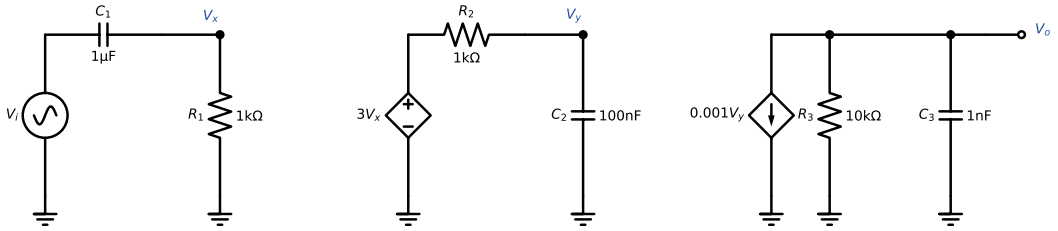
$$|H(j\omega_{in})| = 4.23, \qquad \angle H(j\omega_{in}) = -0.561 \text{ rad}$$
$$v_{o,ac}(t) = 4.23 \times 0.2 \sin(2\pi \cdot 20000\,t - 0.561) = 0.847 \sin(2\pi \cdot 20000\,t - 0.561)$$

$$\boxed{V_o(t) = 0.847 \sin(2\pi \cdot 20000\,t - 0.561)\ \text{V}}$$

Answer

$$H(s) = 5\,\frac{1}{1+\dfrac{s}{2\pi\cdot 31.8\text{k}}} \quad \text{(corners: 31.8 kHz)}$$
$$V_o(t) = 0.847 \sin(2\pi \cdot 20000\,t - 0.561)\ \text{V}$$

Solution 1C.12 **Medium**



Step 1 — Identify the sources

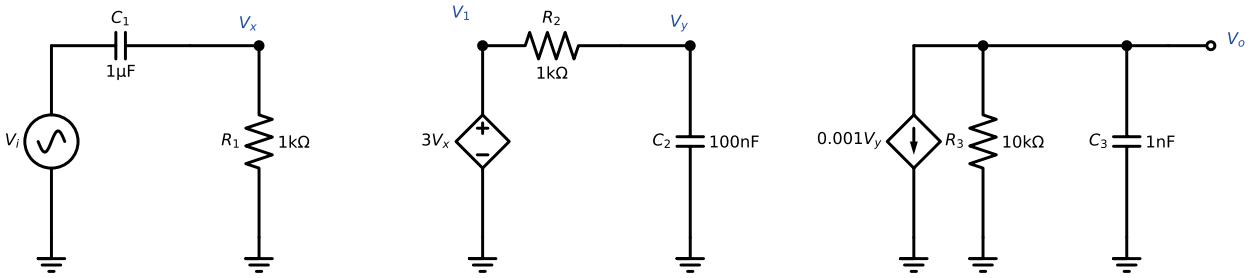
Independent sources: the AC input $V_i(t)$. The diamond-shaped sources are *dependent* sources: they are not inputs, they only copy/scale a voltage that already exists in the circuit, so they stay in the circuit in every step. There is no DC source, so $V_{o,dc} = 0$ and $V_o(t) = v_{o,ac}(t)$: Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — AC circuit

Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL. AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_x \to V_1 \to V_y \to V_o$. The transfer function is the product of the node-to-node ratios (each one is a simple divider, tool T2, or a dependent source, tool T4):

$$H(s) = \frac{V_o}{V_i} = \frac{V_x}{V_i} \cdot \frac{V_1}{V_x} \cdot \frac{V_y}{V_1} \cdot \frac{V_o}{V_y}$$

Impedances seen *looking right* from a node (everything from that node to ground, excluding what is on its left). Compute them from the far right toward the input:

$$Z_o = R_3 \parallel \frac{1}{sC_3} = \frac{10^9}{s + 10^5}$$

Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_x}{V_i} = \frac{R_1}{\frac{1}{sC_1} + R_1} = \frac{s}{s + 1000}$$

$$\frac{V_1}{V_x} = 3 \qquad \text{(ideal VCVS: } V_1 = 3V_x\text{)}$$

$$\frac{V_y}{V_1} = \frac{\frac{1}{sC_2}}{R_2 + \frac{1}{sC_2}} = \frac{10000}{s + 10000}$$

$$\frac{V_o}{V_y} = -0.001\, Z_o = \frac{-10^6}{s + 10^5}$$

(VCCS: the current $0.001V_y$ flows out of node V_o and through Z_o to ground, so $V_o = -0.001V_y\, Z_o$ by Ohm's law.) Multiply the links and simplify (common factors cancel):

$$H(s) = \frac{-3 \times 10^{10}\, s}{s^3 + 1.11 \times 10^5\, s^2 + 1.11 \times 10^9\, s + 10^{12}}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Denominator: $s^3 + 1.11 \times 10^5\, s^2 + 1.11 \times 10^9\, s + 10^{12}$; this is degree 3: use a calculator's polynomial solver (or `numpy.roots`) $\Rightarrow s = -1000,\ -10000,\ -100000$.

Factored form (leading coefficient in front):

$$H(s) = -3 \times 10^{10}\, \frac{s}{(s + 1000)(s + 10000)(s + 10^5)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = -30\, \frac{s}{s + 2\pi \cdot 159}\, \frac{1}{1 + \frac{s}{2\pi \cdot 1.59\text{k}}}\, \frac{1}{1 + \frac{s}{2\pi \cdot 15.9\text{k}}}, \qquad K = -30$$

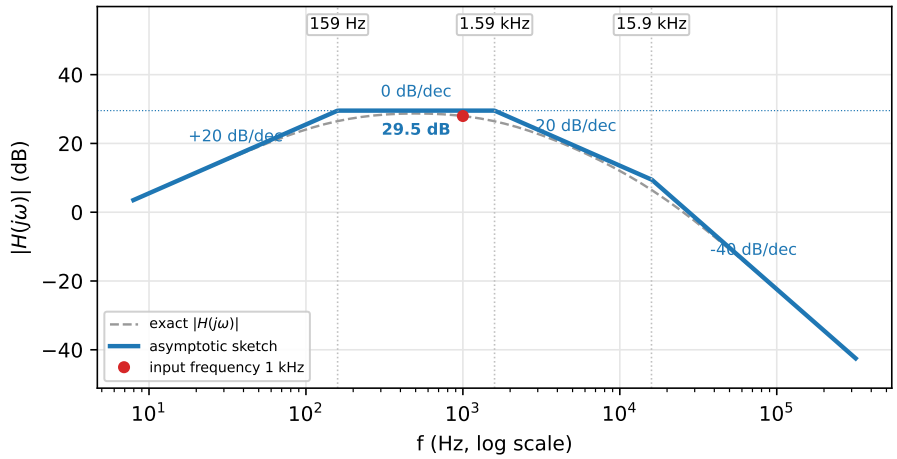
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	1000	159 Hz	−20 dB/dec
pole	10000	1.59 kHz	−20 dB/dec
pole	10^5	15.9 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 0.03\, s$, so the plot starts with a slope of +20 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 159$ Hz	+20 dB/dec	29.5 dB at end
159 Hz – 1.59 kHz	0 dB/dec (flat)	29.5 dB at start, 29.5 dB at end
1.59 kHz – 15.9 kHz	−20 dB/dec	29.5 dB at start, 9.54 dB at end
$f > 15.9$ kHz	−40 dB/dec	9.54 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10}30 = 29.5\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.1\sin(2\pi\cdot 1000\,t)$, so $\omega_{in} = 2\pi\cdot 1000 = 6280\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-30	30	-3.14
$\dfrac{s}{s+2\pi\cdot 159}$	0.9876	0.158
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 1.59\text{k}}}$	0.8467	-0.561
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 15.9\text{k}}}$	0.998	-0.0627
total	**25.04** (product)	$-$**3.61** (sum)

$$|H(j\omega_{in})| = 25, \qquad \angle H(j\omega_{in}) = -3.61 \text{ rad}$$
$$v_{o,ac}(t) = 25\times 0.1\,\sin(2\pi\cdot 1000\,t - 3.61) = 2.5\,\sin(2\pi\cdot 1000\,t - 3.61)$$

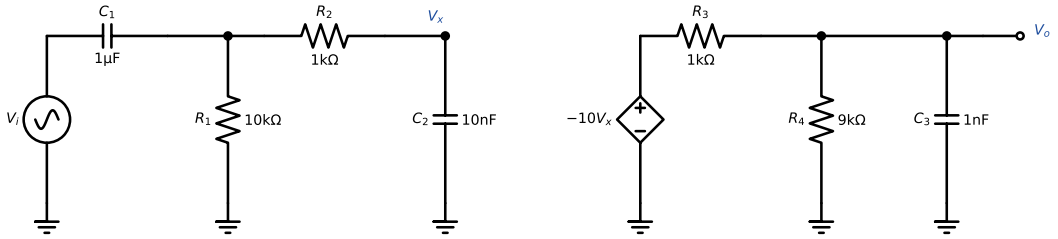
$$\boxed{V_o(t) = 2.5\,\sin(2\pi\cdot 1000\,t - 3.61)\ \text{V}}$$

Answer

$$H(s) = -30\,\frac{s}{s + 2\pi \cdot 159}\,\frac{1}{1 + \dfrac{s}{2\pi \cdot 1.59\text{k}}}\,\frac{1}{1 + \dfrac{s}{2\pi \cdot 15.9\text{k}}} \quad \text{(corners: 159 Hz, 1.59 kHz, 15.9 kHz)}$$

$$V_o(t) = 2.5\,\sin\left(2\pi \cdot 1000\,t - 3.61\right)\ \text{V}$$

Solution 1C.13 **Hard**



Step 1 — Identify the sources

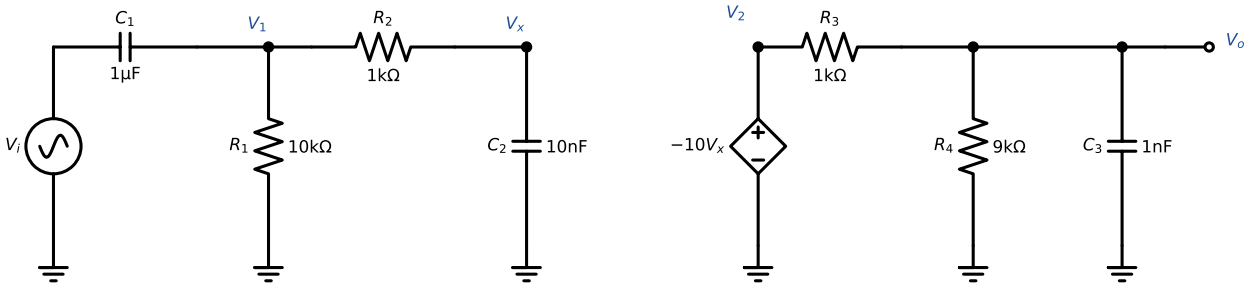
Independent sources: the AC input $V_i(t)$. The diamond-shaped sources are *dependent* sources: they are not inputs, they only copy/scale a voltage that already exists in the circuit, so they stay in the circuit in every step. There is no DC source, so $V_{o,dc} = 0$ and $V_o(t) = v_{o,ac}(t)$: Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — AC circuit

Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL. AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_1 \to V_x \to V_2 \to V_o$. The transfer function is the product of the node-to-node ratios (each one is a simple divider, tool T2, or a dependent source, tool T4):

$$H(s) = \frac{V_o}{V_i} = \frac{V_1}{V_i} \cdot \frac{V_x}{V_1} \cdot \frac{V_2}{V_x} \cdot \frac{V_o}{V_2}$$

Impedances seen *looking right* from a node (everything from that node to ground, excluding what is on its left). Compute them from the far right toward the input:

$$Z_1 = R_1 \parallel \left(R_2 + \frac{1}{sC_2}\right) = \frac{909\,s + 9.09 \times 10^7}{s + 9090}$$

$$Z_o = R_4 \parallel \frac{1}{sC_3} = \frac{10^9}{s + 1.11 \times 10^5}$$

Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_1}{V_i} = \frac{Z_1}{\frac{1}{sC_1} + Z_1} = \frac{s^2 + 10^5\,s}{s^2 + 1.01 \times 10^5\,s + 10^7}$$

$$\frac{V_x}{V_1} = \frac{\frac{1}{sC_2}}{R_2 + \frac{1}{sC_2}} = \frac{10^5}{s + 10^5}$$

$$\frac{V_2}{V_x} = -10 \qquad \text{(ideal VCVS: } V_2 = -10V_x\text{)}$$

$$\frac{V_o}{V_2} = \frac{Z_o}{R_3 + Z_o} = \frac{10^6}{s + 1.11 \times 10^6}$$

Multiply the links and simplify (common factors cancel):

$$H(s) = \frac{-10^{12}\, s}{s^3 + 1.21 \times 10^6\, s^2 + 1.12 \times 10^{11}\, s + 1.11 \times 10^{13}}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Denominator: $s^3 + 1.21 \times 10^6\, s^2 + 1.12 \times 10^{11}\, s + 1.11 \times 10^{13}$; this is degree 3: use a calculator's polynomial solver (or `numpy.roots`) $\Rightarrow s = -99,\ -1.01 \times 10^5,\ -1.11 \times 10^6$.

Factored form (leading coefficient in front):

$$H(s) = -10^{12}\, \frac{s}{(s + 99)(s + 1.01 \times 10^5)(s + 1.11 \times 10^6)}$$

Bode-ready form: rewrite every $(s + p) = p\left(1 + \frac{s}{p}\right)$ and $(s + z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = -8.91\, \frac{s}{s + 2\pi \cdot 15.8}\, \frac{1}{1 + \frac{s}{2\pi \cdot 16.1\text{k}}}\, \frac{1}{1 + \frac{s}{2\pi \cdot 177\text{k}}}, \qquad K = -8.91$$

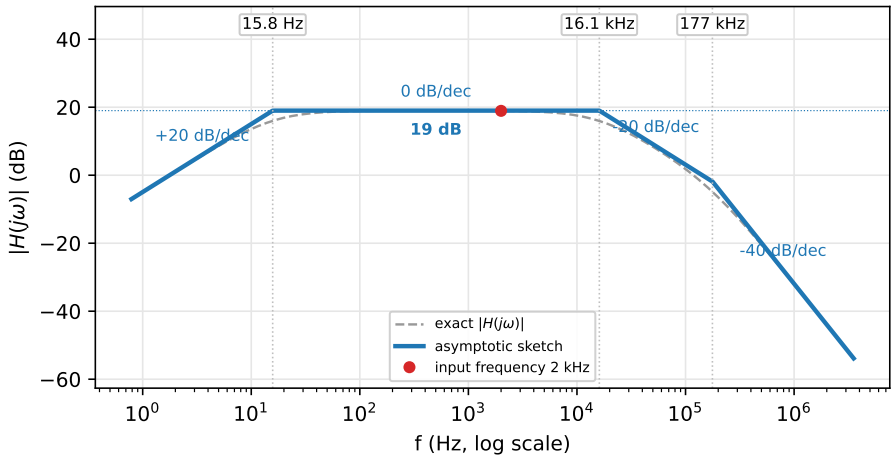
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	99	15.8 Hz	−20 dB/dec
pole	1.01×10^5	16.1 kHz	−20 dB/dec
pole	1.11×10^6	177 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 0.09\, s$, so the plot starts with a slope of +20 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 15.8$ Hz	+20 dB/dec	19 dB at end
15.8 Hz – 16.1 kHz	0 dB/dec (flat)	19 dB at start, 19 dB at end
16.1 kHz – 177 kHz	−20 dB/dec	19 dB at start, -1.83 dB at end
$f > 177$ kHz	−40 dB/dec	-1.83 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10} 8.91 = 19\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.05\sin(2\pi\cdot 2000\,t)$, so $\omega_{in} = 2\pi\cdot 2000 = 12600\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-8.91	8.911	-3.14
$\dfrac{s}{s+2\pi\cdot 15.8}$	1	0.00788
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 16.1\text{k}}}$	0.9923	-0.124
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 177\text{k}}}$	0.9999	-0.0113
total	**8.842** (product)	$-$**3.27** (sum)

$$|H(j\omega_{in})| = 8.84, \qquad \angle H(j\omega_{in}) = -3.27 \text{ rad}$$
$$v_{o,ac}(t) = 8.84\times 0.05\,\sin(2\pi\cdot 2000\,t - 3.27) = 0.442\,\sin(2\pi\cdot 2000\,t - 3.27)$$

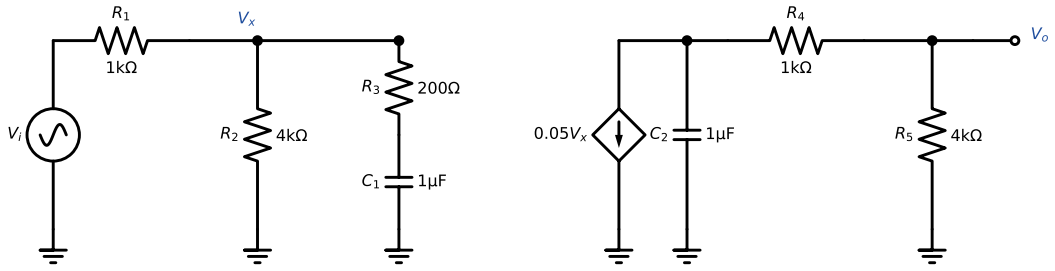
$$\boxed{V_o(t) = 0.442\,\sin(2\pi\cdot 2000\,t - 3.27)\ \text{V}}$$

Answer

$$H(s) = -8.91\,\frac{s}{s + 2\pi \cdot 15.8}\,\frac{1}{1 + \dfrac{s}{2\pi \cdot 16.1\text{k}}}\,\frac{1}{1 + \dfrac{s}{2\pi \cdot 177\text{k}}} \quad \text{(corners: 15.8 Hz, 16.1 kHz, 177 kHz)}$$

$$V_o(t) = 0.442\,\sin\left(2\pi \cdot 2000\,t - 3.27\right)\ \text{V}$$

Solution 1C.14 **Hard**



Step 1 — Identify the sources

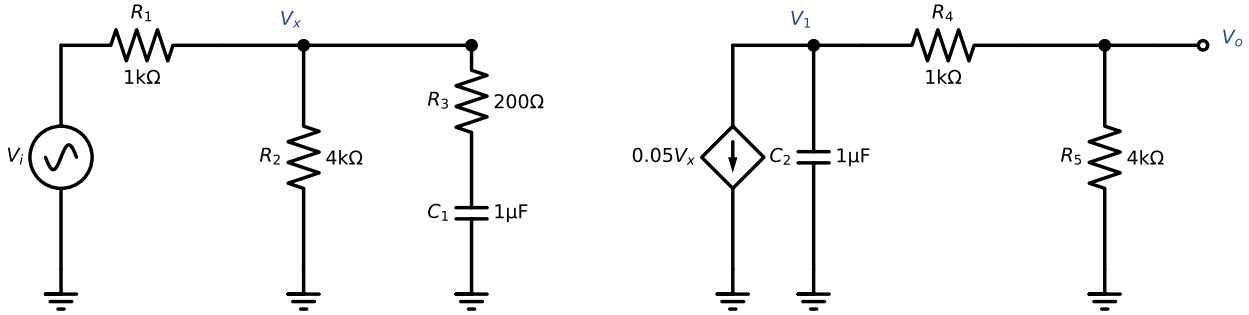
Independent sources: the AC input $V_i(t)$. The diamond-shaped sources are *dependent* sources: they are not inputs, they only copy/scale a voltage that already exists in the circuit, so they stay in the circuit in every step. There is no DC source, so $V_{o,dc} = 0$ and $V_o(t) = v_{o,ac}(t)$: Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — AC circuit

Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL. AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_x \to V_1 \to V_o$. The transfer function is the product of the node-to-node ratios (each one is a simple divider, tool T2, or a dependent source, tool T4):

$$H(s) = \frac{V_o}{V_i} = \frac{V_x}{V_i} \cdot \frac{V_1}{V_x} \cdot \frac{V_o}{V_1}$$

Impedances seen *looking right* from a node (everything from that node to ground, excluding what is on its left). Compute them from the far right toward the input:

$$Z_x = R_2 \parallel \left(R_3 + \frac{1}{sC_1}\right) = \frac{190\,s + 9.52 \times 10^5}{s + 238}$$

$$Z_1 = \frac{1}{sC_2} \parallel (R_4 + R_5) = \frac{10^6}{s + 200}$$

Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with

a plain s^n.)

$$\frac{V_x}{V_i} = \frac{Z_x}{R_1 + Z_x} = \frac{0.16\,s + 800}{s + 1000}$$
$$\frac{V_1}{V_x} = -0.05\,Z_1 = \frac{-50000}{s + 200}$$
$$\frac{V_o}{V_1} = \frac{R_5}{R_4 + R_5} = 0.8$$

(VCCS: the current $0.05V_x$ flows out of node V_1 and through Z_1 to ground, so $V_1 = -0.05V_x\,Z_1$ by Ohm's law.) Multiply the links and simplify (common factors cancel):

$$H(s) = \frac{-6400\,s - 3.2 \times 10^7}{s^2 + 1200\,s + 2 \times 10^5}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: $-6400\,s - 3.2 \times 10^7$; divide by the leading coefficient -6400 (this does not move the roots): $s + 5000$; $s + 5000 = 0 \Rightarrow s = -5000$.
- Denominator: $s^2 + 1200\,s + 2 \times 10^5$; quadratic formula:

$$s = \frac{-1200 \pm \sqrt{(1200)^2 - 4(2 \times 10^5)}}{2} = \frac{-1200 \pm 800}{2} \Rightarrow s = -200,\ -1000$$

Factored form (leading coefficient in front):

$$H(s) = -6400\,\frac{(s + 5000)}{(s + 200)(s + 1000)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

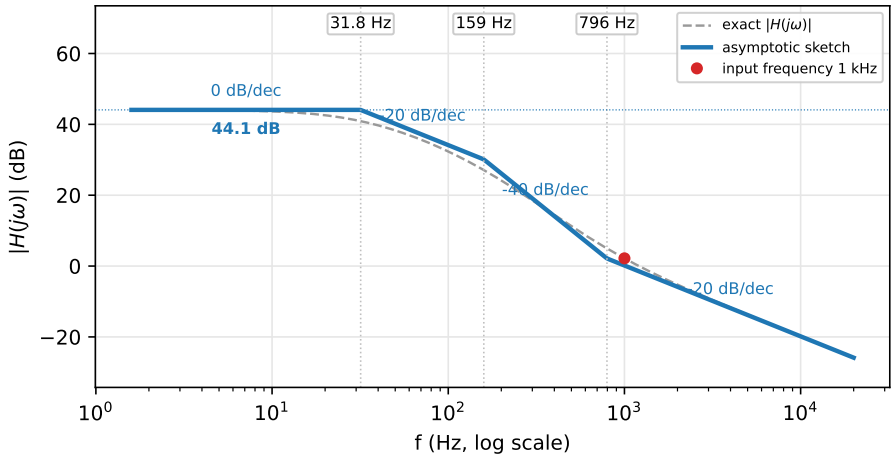
$$H(s) = -160\,\frac{1}{1 + \dfrac{s}{2\pi \cdot 31.8}}\,\frac{1}{1 + \dfrac{s}{2\pi \cdot 159}}\left(1 + \frac{s}{2\pi \cdot 796}\right), \qquad K = -160$$

type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	200	31.8 Hz	−20 dB/dec
pole	1000	159 Hz	−20 dB/dec
zero	5000	796 Hz	+20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 160 = 44.1$ dB (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 31.8$ Hz	0 dB/dec (flat)	44.1 dB at end
31.8 Hz – 159 Hz	−20 dB/dec	44.1 dB at start, 30.1 dB at end
159 Hz – 796 Hz	−40 dB/dec	30.1 dB at start, 2.14 dB at end
$f > 796$ Hz	−20 dB/dec	2.14 dB at start



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.02\sin(2\pi \cdot 1000\,t)$, so $\omega_{in} = 2\pi \cdot 1000 = 6280$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-160	160	-3.14
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 31.8}}$	0.03181	-1.54
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 159}}$	0.1572	-1.41
$\left(1+\dfrac{s}{2\pi \cdot 796}\right)$	1.606	0.899
total	**1.285** (product)	**−5.19** (sum)

$$|H(j\omega_{in})| = 1.28, \qquad \angle H(j\omega_{in}) = -5.19 \text{ rad}$$
$$v_{o,ac}(t) = 1.28 \times 0.02 \sin(2\pi \cdot 1000\,t - 5.19) = 0.0257 \sin(2\pi \cdot 1000\,t - 5.19)$$

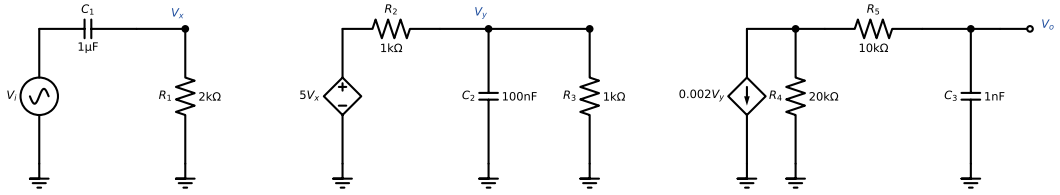
$$\boxed{V_o(t) = 0.0257 \sin\left(2\pi \cdot 1000\, t - 5.19\right)\ \text{V}}$$

Answer

$$H(s) = -160\, \frac{1}{1 + \dfrac{s}{2\pi \cdot 31.8}}\, \frac{1}{1 + \dfrac{s}{2\pi \cdot 159}}\, \left(1 + \frac{s}{2\pi \cdot 796}\right) \quad \text{(corners: 31.8 Hz, 159 Hz, 796 Hz)}$$

$$V_o(t) = 0.0257 \sin\left(2\pi \cdot 1000\, t - 5.19\right)\ \text{V}$$

Solution 1C.15 **Hard**



Step 1 — Identify the sources

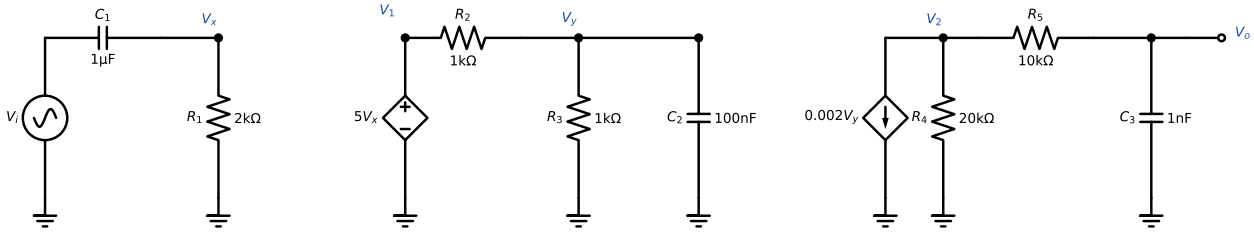
Independent sources: the AC input $V_i(t)$. The diamond-shaped sources are *dependent* sources: they are not inputs, they only copy/scale a voltage that already exists in the circuit, so they stay in the circuit in every step. There is no DC source, so $V_{o,dc} = 0$ and $V_o(t) = v_{o,ac}(t)$: Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — AC circuit

Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL. AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_x \to V_1 \to V_y \to V_2 \to V_o$. The transfer function is the product of the node-to-node ratios (each one is a simple divider, tool T2, or a dependent source, tool T4):

$$H(s) = \frac{V_o}{V_i} = \frac{V_x}{V_i} \cdot \frac{V_1}{V_x} \cdot \frac{V_y}{V_1} \cdot \frac{V_2}{V_y} \cdot \frac{V_o}{V_2}$$

Impedances seen *looking right* from a node (everything from that node to ground, excluding what is on its left). Compute them from the far right toward the input:

$$Z_y = R_3 \parallel \frac{1}{sC_2} = \frac{10^7}{s + 10000}$$

$$Z_2 = R_4 \parallel \left(R_5 + \frac{1}{sC_3}\right) = \frac{6670\,s + 6.67 \times 10^8}{s + 33300}$$

Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_x}{V_i} = \frac{R_1}{\frac{1}{sC_1} + R_1} = \frac{s}{s + 500}$$

$$\frac{V_1}{V_x} = 5 \qquad \text{(ideal VCVS: } V_1 = 5V_x\text{)}$$
$$\frac{V_y}{V_1} = \frac{Z_y}{R_2 + Z_y} = \frac{10000}{s + 20000}$$
$$\frac{V_2}{V_y} = -0.002\, Z_2 = \frac{-13.3\, s - 1.33 \times 10^6}{s + 33300}$$
$$\frac{V_o}{V_2} = \frac{\frac{1}{sC_3}}{R_5 + \frac{1}{sC_3}} = \frac{10^5}{s + 10^5}$$

(VCCS: the current $0.002V_y$ flows out of node V_2 and through Z_2 to ground, so $V_2 = -0.002V_y\, Z_2$ by Ohm's law.) Multiply the links and simplify (common factors cancel):

$$H(s) = \frac{-6.67 \times 10^{10}\, s}{s^3 + 53800\, s^2 + 6.93 \times 10^8\, s + 3.33 \times 10^{11}}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Denominator: $s^3 + 53800\, s^2 + 6.93 \times 10^8\, s + 3.33 \times 10^{11}$; this is degree 3: use a calculator's polynomial solver (or `numpy.roots`) $\Rightarrow s = -500,\ -20000,\ -33300$.

Factored form (leading coefficient in front):

$$H(s) = -6.67 \times 10^{10}\, \frac{s}{(s + 500)(s + 20000)(s + 33300)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = -100\, \frac{s}{s + 2\pi \cdot 79.6}\, \frac{1}{1 + \frac{s}{2\pi \cdot 3.18\text{k}}}\, \frac{1}{1 + \frac{s}{2\pi \cdot 5.31\text{k}}}, \qquad K = -100$$

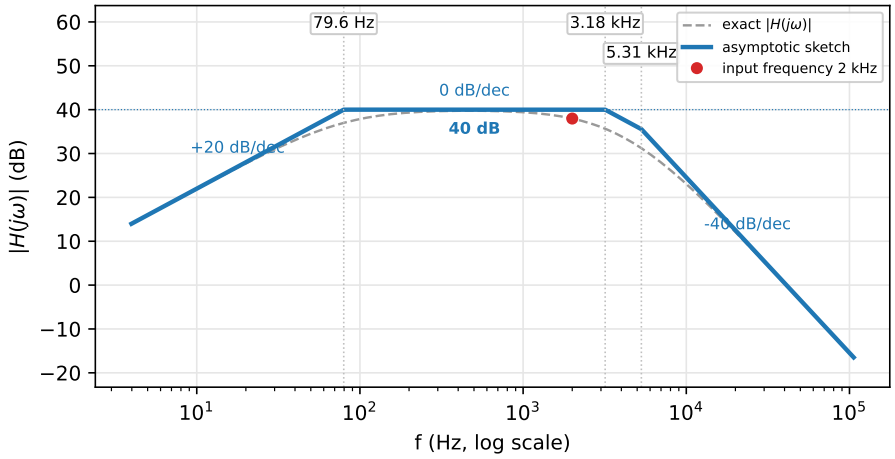
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	500	79.6 Hz	−20 dB/dec
pole	20000	3.18 kHz	−20 dB/dec
pole	33300	5.31 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 0.2\, s$, so the plot starts with a slope of +20 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 79.6$ Hz	+20 dB/dec	40 dB at end
79.6 Hz – 3.18 kHz	0 dB/dec (flat)	40 dB at start, 40 dB at end
3.18 kHz – 5.31 kHz	−20 dB/dec	40 dB at start, 35.6 dB at end
$f > 5.31$ kHz	−40 dB/dec	35.6 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10}100 = 40\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.1\sin\left(2\pi\cdot 2000\,t + \frac{\pi}{4}\right)$, so $\omega_{in} = 2\pi\cdot 2000 = 12600\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-100	100	-3.14
$\dfrac{s}{s+2\pi\cdot 79.6}$	0.9992	0.0398
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 3.18\text{k}}}$	0.8467	-0.561
$\dfrac{1}{1+\dfrac{s}{2\pi\cdot 5.31\text{k}}}$	0.9357	-0.361
total	**79.17** (product)	**−4.02** (sum)

$$|H(j\omega_{in})| = 79.2, \qquad \angle H(j\omega_{in}) = -4.02 \text{ rad}$$

$$v_{o,ac}(t) = 79.2 \times 0.1 \sin(2\pi \cdot 2000\,t - 3.24) = 7.92 \sin(2\pi \cdot 2000\,t - 3.24)$$

(phase: $\theta + \angle H = \frac{\pi}{4} + (-4.02) = -3.24\,\text{rad.}$)

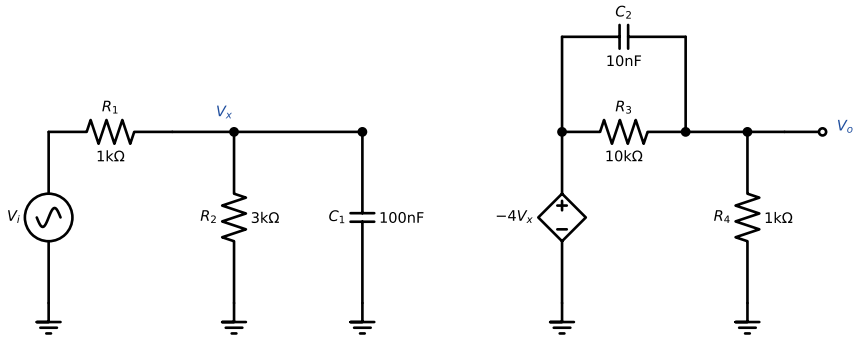
$$\boxed{V_o(t) = 7.92 \sin(2\pi \cdot 2000\,t - 3.24)\ \text{V}}$$

Answer

$$H(s) = -100\,\frac{s}{s + 2\pi \cdot 79.6}\,\frac{1}{1 + \dfrac{s}{2\pi \cdot 3.18\text{k}}}\,\frac{1}{1 + \dfrac{s}{2\pi \cdot 5.31\text{k}}} \quad \text{(corners: 79.6 Hz, 3.18 kHz, 5.31 kHz)}$$

$$V_o(t) = 7.92 \sin(2\pi \cdot 2000\,t - 3.24)\ \text{V}$$

Solution 1C.16 **Hard**



Step 1 — Identify the sources

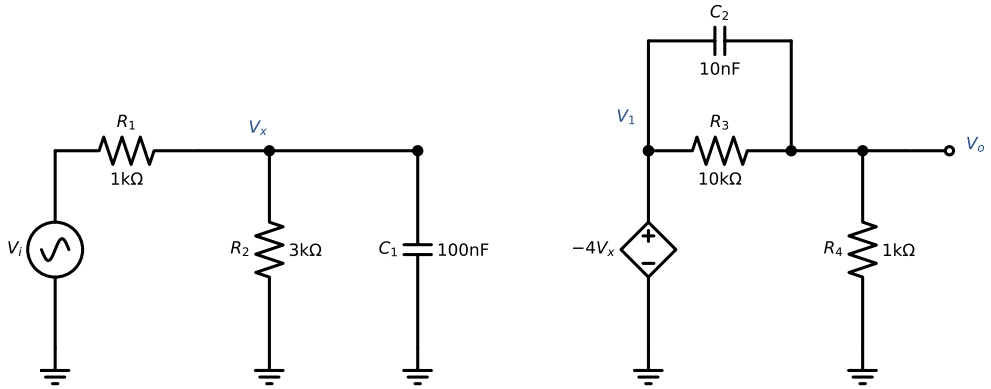
Independent sources: the AC input $V_i(t)$. The diamond-shaped sources are *dependent* sources: they are not inputs, they only copy/scale a voltage that already exists in the circuit, so they stay in the circuit in every step. There is no DC source, so $V_{o,dc} = 0$ and $V_o(t) = v_{o,ac}(t)$: Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — AC circuit

Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL. AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_x \to V_1 \to V_o$. The transfer function is the product of the node-to-node ratios (each one is a simple divider, tool T2, or a dependent source, tool T4):

$$H(s) = \frac{V_o}{V_i} = \frac{V_x}{V_i} \cdot \frac{V_1}{V_x} \cdot \frac{V_o}{V_1}$$

Impedances seen *looking right* from a node (everything from that node to ground, excluding what is on its left). Compute them from the far right toward the input:

$$Z_x = R_2 \parallel \frac{1}{sC_1} = \frac{10^7}{s + 3330}$$

Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_x}{V_i} = \frac{Z_x}{R_1 + Z_x} = \frac{10000}{s + 13300}$$
$$\frac{V_1}{V_x} = -4 \qquad \text{(ideal VCVS: } V_1 = -4V_x\text{)}$$
$$\frac{V_o}{V_1} = \frac{R_4}{R_3 \parallel \frac{1}{sC_2} + R_4} = \frac{s + 10000}{s + 1.1 \times 10^5}$$

Multiply the links and simplify (common factors cancel):

$$H(s) = \frac{-40000\, s - 4 \times 10^8}{s^2 + 1.23 \times 10^5\, s + 1.47 \times 10^9}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: $-40000\, s - 4 \times 10^8$; divide by the leading coefficient -40000 (this does not move the roots): $s + 10000$; $s + 10000 = 0 \Rightarrow s = -10000$.
- Denominator: $s^2 + 1.23 \times 10^5\, s + 1.47 \times 10^9$; quadratic formula:

$$s = \frac{-1.23 \times 10^5 \pm \sqrt{(1.23 \times 10^5)^2 - 4(1.47 \times 10^9)}}{2} = \frac{-1.23 \times 10^5 \pm 96700}{2} \Rightarrow s = -13300,\ -1.1 \times 10^5$$

Factored form (leading coefficient in front):

$$H(s) = -40000\, \frac{(s + 10000)}{(s + 13300)(s + 1.1 \times 10^5)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Then every factor except K is ≈ 1 inside the pass-band:

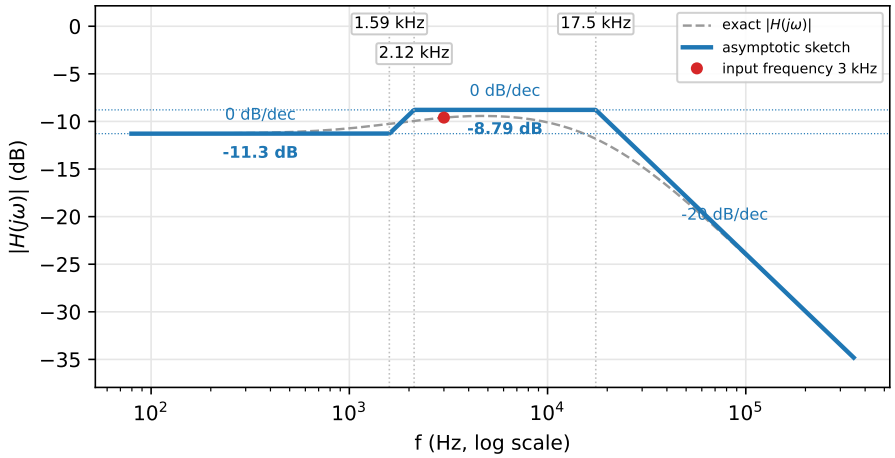
$$H(s) = -0.273\, \left(1 + \frac{s}{2\pi \cdot 1.59\text{k}}\right)\, \frac{1}{1 + \frac{s}{2\pi \cdot 2.12\text{k}}}\, \frac{1}{1 + \frac{s}{2\pi \cdot 17.5\text{k}}}, \qquad K = -0.273$$

type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
zero	10000	1.59 kHz	+20 dB/dec
pole	13300	2.12 kHz	−20 dB/dec
pole	1.1×10^5	17.5 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner, all factors are $\approx$ constant: $|H| \approx 0.273 = -11.3\,\text{dB}$ (flat, 0 dB/dec). Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 1.59$ kHz	0 dB/dec (flat)	-11.3 dB at end
1.59 kHz – 2.12 kHz	+20 dB/dec	-11.3 dB at start, -8.79 dB at end
2.12 kHz – 17.5 kHz	0 dB/dec (flat)	-8.79 dB at start, -8.79 dB at end
$f > 17.5$ kHz	−20 dB/dec	-8.79 dB at start



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.1\cos(2\pi \cdot 3000\,t)$, so $\omega_{in} = 2\pi \cdot 3000 = 18800$ rad/s. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
-0.273	0.2727	-3.14
$\left(1+\dfrac{s}{2\pi \cdot 1.59\text{k}}\right)$	2.134	1.08
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 2.12\text{k}}}$	0.5775	-0.955
$\dfrac{1}{1+\dfrac{s}{2\pi \cdot 17.5\text{k}}}$	0.9856	-0.17
total	**0.3312** (product)	$-$**3.18** (sum)

$$|H(j\omega_{in})| = 0.331, \qquad \angle H(j\omega_{in}) = -3.18 \text{ rad}$$
$$v_{o,ac}(t) = 0.331 \times 0.1\cos(2\pi \cdot 3000\,t - 3.18) = 0.0331\cos(2\pi \cdot 3000\,t - 3.18)$$

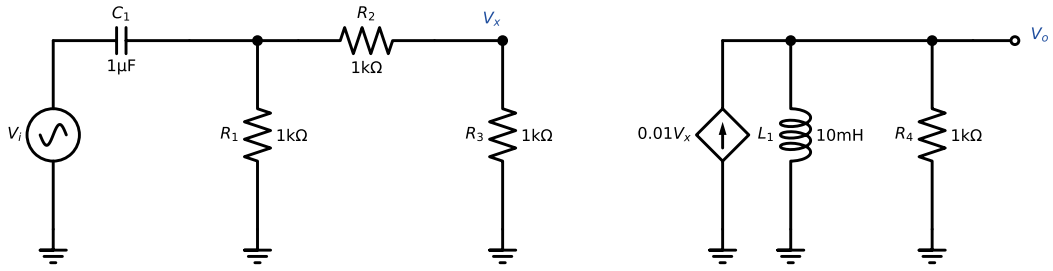
$$\boxed{V_o(t) = 0.0331\ \cos\left(2\pi \cdot 3000\, t - 3.18\right)\ \text{V}}$$

Answer

$H(s) = -0.273\ \left(1 + \dfrac{s}{2\pi \cdot 1.59\text{k}}\right)\ \dfrac{1}{1 + \dfrac{s}{2\pi \cdot 2.12\text{k}}}\ \dfrac{1}{1 + \dfrac{s}{2\pi \cdot 17.5\text{k}}}$ (corners: 1.59 kHz, 2.12 kHz, 17.5 kHz)

$V_o(t) = 0.0331\ \cos\left(2\pi \cdot 3000\, t - 3.18\right)\ \text{V}$

Solution 1C.17 **Hard**



Step 1 — Identify the sources

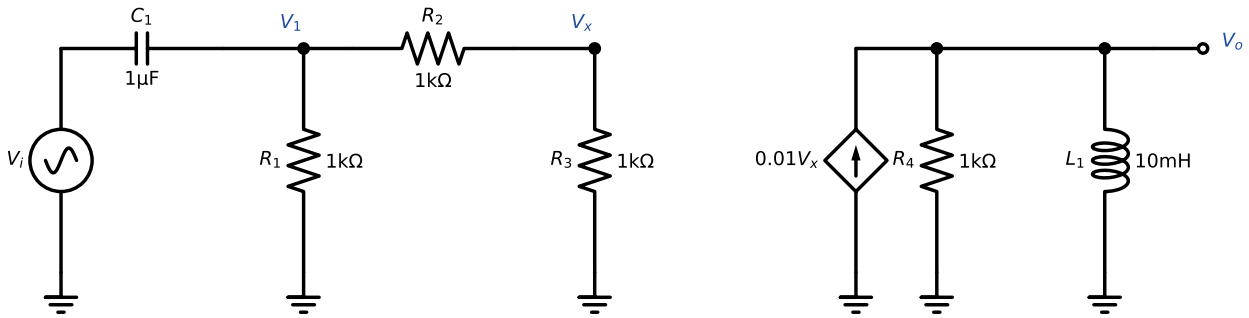
Independent sources: the AC input $V_i(t)$. The diamond-shaped sources are *dependent* sources: they are not inputs, they only copy/scale a voltage that already exists in the circuit, so they stay in the circuit in every step. There is no DC source, so $V_{o,dc} = 0$ and $V_o(t) = v_{o,ac}(t)$: Step 2 is skipped.

Step 2 — DC circuit

No DC sources: $V_{o,dc} = 0$.

Step 3 — AC circuit

Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL. AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_1 \to V_x \to V_o$. The transfer function is the product of the node-to-node ratios (each one is a simple divider, tool T2, or a dependent source, tool T4):

$$H(s) = \frac{V_o}{V_i} = \frac{V_1}{V_i} \cdot \frac{V_x}{V_1} \cdot \frac{V_o}{V_x}$$

Impedances seen *looking right* from a node (everything from that node to ground, excluding what is on its left). Compute them from the far right toward the input:

$$Z_1 = R_1 \parallel (R_2 + R_3) = 667$$
$$Z_o = R_4 \parallel sL_1 = \frac{1000\,s}{s + 10^5}$$

Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with

a plain s^n.)

$$\frac{V_1}{V_i} = \frac{Z_1}{\frac{1}{sC_1} + Z_1} = \frac{s}{s + 1500}$$
$$\frac{V_x}{V_1} = \frac{R_3}{R_2 + R_3} = 0.5$$
$$\frac{V_o}{V_x} = 0.01\, Z_o = \frac{10\, s}{s + 10^5}$$

(VCCS: the current $0.01V_x$ flows into node V_o and through Z_o to ground, so $V_o = 0.01V_x\, Z_o$ by Ohm's law.) Multiply the links and simplify (common factors cancel):

$$H(s) = \frac{5\, s^2}{s^2 + 1.01 \times 10^5\, s + 1.5 \times 10^8}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s^2 \Rightarrow$ 2 zeros at $s = 0$.
- Denominator: $s^2 + 1.01 \times 10^5\, s + 1.5 \times 10^8$; quadratic formula:

$$s = \frac{-1.01 \times 10^5 \pm \sqrt{(1.01 \times 10^5)^2 - 4(1.5 \times 10^8)}}{2} = \frac{-1.01 \times 10^5 \pm 98500}{2} \Rightarrow s = -1500,\ -10^5$$

Factored form (leading coefficient in front):

$$H(s) = 5\, \frac{s^2}{(s + 1500)(s + 10^5)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

$$H(s) = 5\, \frac{s}{s + 2\pi \cdot 239}\, \frac{s}{s + 2\pi \cdot 15.9\text{k}}, \qquad K = 5$$

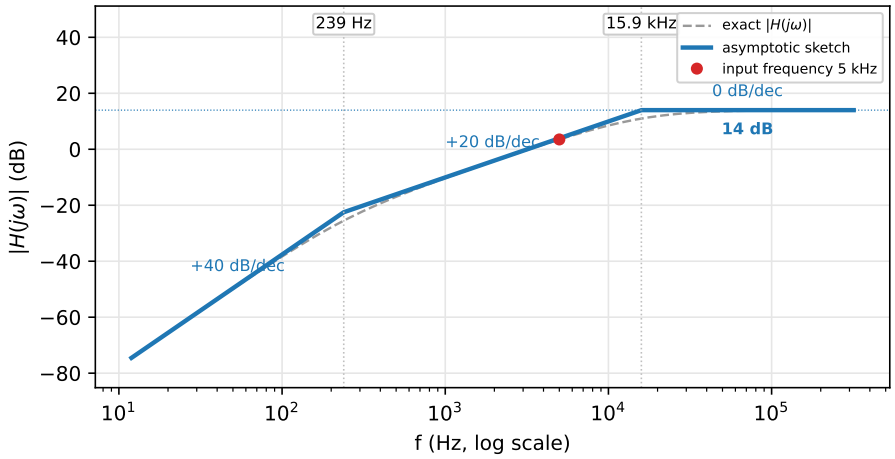
type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	1500	239 Hz	−20 dB/dec
pole	10^5	15.9 kHz	−20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 3.33 \times 10^{-8}\, s^2$, so the plot starts with a slope of +40 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f <$ 239 Hz	+40 dB/dec	-22.5 dB at end
239 Hz – 15.9 kHz	+20 dB/dec	-22.5 dB at start, 14 dB at end
$f >$ 15.9 kHz	0 dB/dec (flat)	14 dB at start

The flat (pass-band) level is $20\log_{10}|K| = 20\log_{10}5 = 14\,\text{dB}$.



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.2\sin\left(2\pi\cdot 5000\,t - \frac{\pi}{6}\right)$, so $\omega_{in} = 2\pi\cdot 5000 = 31400\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
5	5	0
$\dfrac{s}{s+2\pi\cdot 239}$	0.9989	0.0477
$\dfrac{s}{s+2\pi\cdot 15.9\text{k}}$	0.2997	1.27
total	**1.497** (product)	**1.31** (sum)

$$|H(j\omega_{in})| = 1.5, \qquad \angle H(j\omega_{in}) = 1.31 \text{ rad}$$
$$v_{o,ac}(t) = 1.5\times 0.2\sin\left(2\pi\cdot 5000\,t + 0.791\right) = 0.299\sin\left(2\pi\cdot 5000\,t + 0.791\right)$$

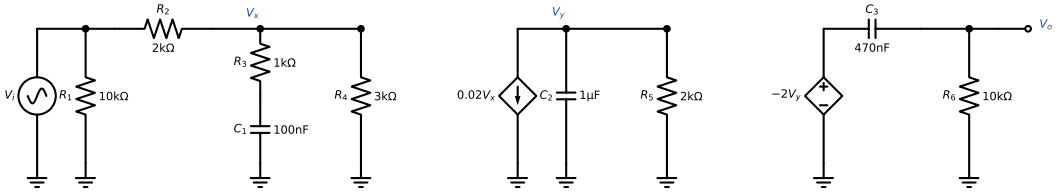
(phase: $\theta + \angle H = -\frac{\pi}{6} + (1.31) = 0.791\,\text{rad}$.)

$$\boxed{V_o(t) = 0.299\sin\left(2\pi\cdot 5000\,t + 0.791\right)\ \text{V}}$$

Answer

$$H(s) = 5\,\frac{s}{s+2\pi\cdot 239}\,\frac{s}{s+2\pi\cdot 15.9\text{k}} \quad \text{(corners: 239 Hz, 15.9 kHz)}$$
$$V_o(t) = 0.299\sin\left(2\pi\cdot 5000\,t + 0.791\right)\ \text{V}$$

Solution 1C.18 **Hard**



Step 1 — Identify the sources

Independent sources: the AC input $V_i(t)$. The diamond-shaped sources are *dependent* sources: they are not inputs, they only copy/scale a voltage that already exists in the circuit, so they stay in the circuit in every step. There is no DC source, so $V_{o,dc} = 0$ and $V_o(t) = v_{o,ac}(t)$: Step 2 is skipped.

Step 2 — DC circuit

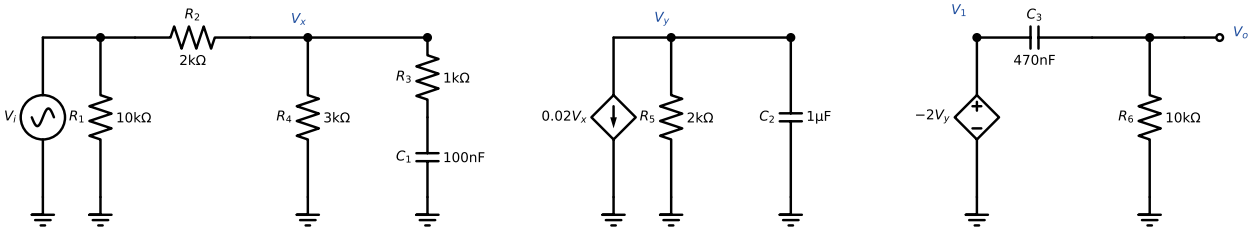
No DC sources: $V_{o,dc} = 0$.

Step 3 — AC circuit

Replace every capacitor by $\frac{1}{sC}$ and every inductor by sL.

- R_1 sits directly across the ideal input source V_i. An ideal voltage source keeps its voltage whatever you connect to it, so R_1 cannot change any node voltage: it drops out of $H(s)$.

AC equivalent circuit (nodes named for the next step):



Step 4 — Transfer function by a chain of dividers

Walk from the input to the output through the named nodes: $V_i \to V_x \to V_y \to V_1 \to V_o$. The transfer function is the product of the node-to-node ratios (each one is a simple divider, tool T2, or a dependent source, tool T4):

$$H(s) = \frac{V_o}{V_i} = \frac{V_x}{V_i} \cdot \frac{V_y}{V_x} \cdot \frac{V_1}{V_y} \cdot \frac{V_o}{V_1}$$

Impedances seen *looking right* from a node (everything from that node to ground, excluding what is on its left). Compute them from the far right toward the input:

$$Z_x = R_4 \parallel \left(R_3 + \frac{1}{sC_1}\right) = \frac{750\,s + 7.5 \times 10^6}{s + 2500}$$

$$Z_y = R_5 \parallel \frac{1}{sC_2} = \frac{10^6}{s + 500}$$

Now each link. (Algebra tip: to simplify a fraction that contains fractions, multiply its top and bottom by every inner denominator; then divide top and bottom by the leading coefficient of the denominator so that it starts with a plain s^n.)

$$\frac{V_x}{V_i} = \frac{Z_x}{R_2 + Z_x} = \frac{0.273\,s + 2730}{s + 4550}$$

$$\frac{V_y}{V_x} = -0.02\, Z_y = \frac{-20000}{s+500}$$
$$\frac{V_1}{V_y} = -2 \qquad \text{(ideal VCVS: } V_1 = -2V_y\text{)}$$
$$\frac{V_o}{V_1} = \frac{R_6}{\frac{1}{sC_3} + R_6} = \frac{s}{s+213}$$

(VCCS: the current $0.02V_x$ flows out of node V_y and through Z_y to ground, so $V_y = -0.02V_x\, Z_y$ by Ohm's law.) Multiply the links and simplify (common factors cancel):

$$H(s) = \frac{10900\, s^2 + 1.09 \times 10^8\, s}{s^3 + 5260\, s^2 + 3.35 \times 10^6\, s + 4.84 \times 10^8}$$

Step 5 — Standard (Bode-ready) form

Find the roots. Every root $s = -p$ of the denominator is a *pole*, every root $s = -z$ of the numerator is a *zero*; $|p|$ and $|z|$ are the corner frequencies (rad/s).

- Numerator: it has the factor $s \Rightarrow 1$ zero at $s = 0$.
- Numerator (after taking out the s factor): $10900\, s + 1.09 \times 10^8$; divide by the leading coefficient 10900 (this does not move the roots): $s + 10000$; $s + 10000 = 0 \Rightarrow s = -10000$.
- Denominator: $s^3 + 5260\, s^2 + 3.35 \times 10^6\, s + 4.84 \times 10^8$; this is degree 3: use a calculator's polynomial solver (or `numpy.roots`) $\Rightarrow s = -213,\ -500,\ -4550$.

Factored form (leading coefficient in front):

$$H(s) = 10900\, \frac{s(s+10000)}{(s+213)(s+500)(s+4550)}$$

Bode-ready form: rewrite every $(s+p) = p\left(1 + \frac{s}{p}\right)$ and $(s+z) = z\left(1 + \frac{s}{z}\right)$, and collect all constants into K. Each s in the numerator is paired with one of the *lowest* poles as $\frac{s}{s+p}$ (that factor is ≈ 1 above p). Then every factor except K is ≈ 1 inside the pass-band:

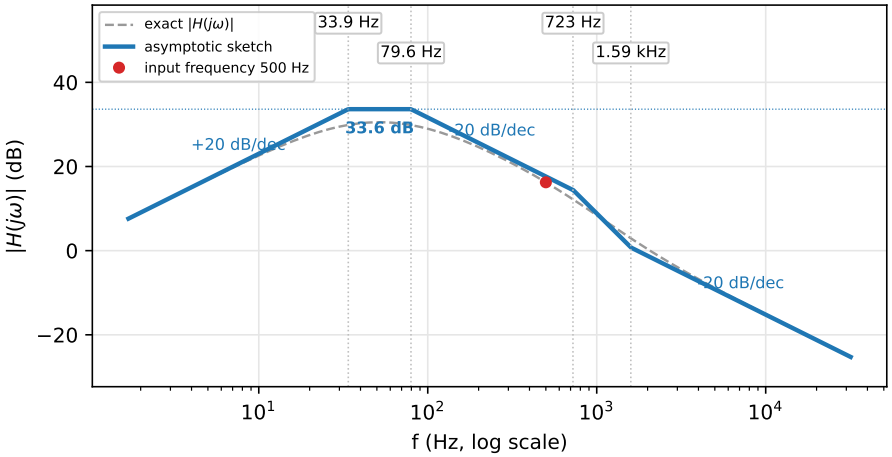
$$H(s) = 48\, \frac{s}{s + 2\pi \cdot 33.9}\, \frac{1}{1 + \frac{s}{2\pi \cdot 79.6}}\, \frac{1}{1 + \frac{s}{2\pi \cdot 723}}\, \left(1 + \frac{s}{2\pi \cdot 1.59\text{k}}\right), \qquad K = 48$$

type	ω_c (rad/s)	$f_c = \omega_c/2\pi$	slope change
pole	213	33.9 Hz	−20 dB/dec
pole	500	79.6 Hz	−20 dB/dec
pole	4550	723 Hz	−20 dB/dec
zero	10000	1.59 kHz	+20 dB/dec

Step 6 — Magnitude Bode plot

Below every corner $|H(s)| \approx 0.226\, s$, so the plot starts with a slope of +20 dB/dec. Then, moving right, each pole bends the line down by 20 dB/dec and each zero bends it up by 20 dB/dec (tool T6):

frequency range	slope	asymptote level
$f < 33.9$ Hz	+20 dB/dec	33.6 dB at end
33.9 Hz – 79.6 Hz	0 dB/dec (flat)	33.6 dB at start, 33.6 dB at end
79.6 Hz – 723 Hz	−20 dB/dec	33.6 dB at start, 14.5 dB at end
723 Hz – 1.59 kHz	−40 dB/dec	14.5 dB at start, 0.756 dB at end
$f > 1.59$ kHz	−20 dB/dec	0.756 dB at start



Blue: the straight-line sketch you are expected to draw. Dashed grey: the exact curve (about 3 dB below the corner of an isolated pole). Red dot: the input frequency of Step 7.

Step 7 — Steady-state output

The input is $V_i(t) = 0.01\cos\left(2\pi \cdot 500\,t + \frac{\pi}{3}\right)$, so $\omega_{in} = 2\pi \cdot 500 = 3140\,\text{rad/s}$. Put $s = j\omega_{in}$ in each factor (tool T5): magnitudes multiply, angles add. Rules: $\left|\frac{s}{s+p}\right| = \frac{\omega}{\sqrt{\omega^2+p^2}}$, $\angle = \frac{\pi}{2} - \tan^{-1}\frac{\omega}{p}$; $\left|\frac{1}{1+s/p}\right| = \frac{1}{\sqrt{1+(\omega/p)^2}}$, $\angle = -\tan^{-1}\frac{\omega}{p}$; $\left|1+\frac{s}{z}\right| = \sqrt{1+(\omega/z)^2}$, $\angle = \tan^{-1}\frac{\omega}{z}$; a negative K adds $-\pi$.

factor	magnitude	angle (rad)
48	48	0
$\dfrac{s}{s + 2\pi \cdot 33.9}$	0.9977	0.0676
$\dfrac{1}{1 + \dfrac{s}{2\pi \cdot 79.6}}$	0.1572	−1.41
$\dfrac{1}{1 + \dfrac{s}{2\pi \cdot 723}}$	0.8226	−0.605
$\left(1 + \dfrac{s}{2\pi \cdot 1.59\text{k}}\right)$	1.048	0.304
total	**6.491** (product)	**−1.65** (sum)

$$|H(j\omega_{in})| = 6.49, \qquad \angle H(j\omega_{in}) = -1.65 \text{ rad}$$
$$v_{o,ac}(t) = 6.49 \times 0.01 \cos(2\pi \cdot 500\,t - 0.599) = 0.0649 \cos(2\pi \cdot 500\,t - 0.599)$$

(phase: $\theta + \angle H = \frac{\pi}{3} + (-1.65) = -0.599\,\text{rad}$.)

$$\boxed{V_o(t) = 0.0649 \cos(2\pi \cdot 500\,t - 0.599) \text{ V}}$$

Answer

$H(s) = 48 \dfrac{s}{s + 2\pi \cdot 33.9} \dfrac{1}{1 + \dfrac{s}{2\pi \cdot 79.6}} \dfrac{1}{1 + \dfrac{s}{2\pi \cdot 723}} \left(1 + \dfrac{s}{2\pi \cdot 1.59\text{k}}\right)$ (corners: 33.9 Hz, 79.6 Hz, 723 Hz, 1.59 kHz)

$V_o(t) = 0.0649 \cos(2\pi \cdot 500\,t - 0.599)$ V